



SUBCONTRACT AGREEMENT

BETWEEN

Desert Services

“Subcontractor”

and

**LGE CORPORATION, an Arizona corporation d/b/a LGE DESIGN BUILD
“Design-Builder”**

**“PROJECT:” International Furniture Direct
Job #: 01-10604
7839 North 73rd Ave
Glendale, AZ 85303**

**“OWNER:” Chicago International Real Estate, LLC d/b/a International Furniture
Direct
5750 McEver Rd.
Ste. A
Oackwood, GA 30566**

In consideration of the mutual covenants and obligations contained herein, Design-Builder and Subcontractor agrees as set forth herein.

Article I. General

1.1 Definitions.

1.1.1 Terms used in this Agreement shall have the meanings set forth in the Design-Build Agreement unless otherwise provided herein, with the following specific terms defined as follows:

1.1.1.1.1 *Contract Documents* are comprised of the following: (i) this Agreement; (ii) all written modifications, amendments, minor changes and Change Orders to this Agreement; (iii) the Construction Documents; and (iv) the Design-Build Agreement, but only to the extent the Design-Build Agreement relates to the work performed by or to be performed by the Subcontractor. The Contract Documents are intended to be complementary and interpreted in harmony so as to avoid conflict, with words and phrases interpreted consistent with construction and design industry standards. In the event of any inconsistency, conflict, or ambiguity between or among the Contract Documents, the Contract Documents shall take precedence in the order in which they are listed in this Section 1.1.1.1.1.

1.1.1.1.2 *Construction Documents* are the documents consisting of Plans and Specifications, prepared or assembled by Design-Builder in accordance with the Design-Build Agreement for the Project, including, without limitation, those Plans and Specifications listed in the attached Exhibit C which such Plans and Specifications are hereby incorporated herein by reference with the same force and effect as though fully set forth herein.

1.1.1.1.3 *Contract Price* has that meaning ascribed to it in Section 6.1.1 below.

1.1.1.1.4 *Day or Days* shall mean calendar days unless otherwise specifically noted in the Contract Documents.

1.1.1.1.5 *Design-Build Agreement* refers to the contract between Design-Builder and an Owner for the design and construction of the Project and all exhibits, attachments, and other Contract Documents enumerated and incorporated therein.

1.1.1.1.6 *Hazardous Conditions* are any materials, wastes, substances and chemicals deemed to be hazardous under applicable Legal Requirements, or the handling, storage, remediation, or disposal of which are regulated by applicable Legal Requirements.

1.1.1.1.7 *Legal Requirements* are all applicable federal, state and local laws, codes, ordinances, rules, regulations, orders and decrees of any government or quasi-government entity having jurisdiction over the parties, the Project or Site, the practices involved in the Project or Site, or any Work.

1.1.1.1.8 *Agreement* refers to this executed contract between Design-Builder and Subcontractor.

1.1.1.1.9 *Owner* refers to the Owner of the Project.

1.1.1.1.10 *Project Schedule* refers to the schedule setting forth the dates by which the various stages of both the design and construction of a Project which must be performed so as to satisfy Design-Builder's obligations to an Owner as may be revised by Design-Builder from time to time.

1.1.1.1.11 *Site* is the land or premises on which the Project is located.

1.1.1.1.12 *Sub-Subcontractor* is any person or entity retained by Subcontractor as an independent contractor to perform a portion of the Subcontractor's Work and shall include materialmen and suppliers.

1.1.1.1.13 *Substantial Completion or Substantially Complete* is the date on which the Project is sufficiently complete in accordance with the Contract Documents so that Owner can occupy and use the Project for its intended purposes.

1.1.1.1.14 *Work* has that meaning ascribed to it in Section 2.1 below and includes, without limitation, that work described in Subcontractor's Scope of Work exhibit attached hereto as Exhibit A.

1.2 **Basic Purpose.**

1.2.1 Design-Builder has contracted with Owner to provide the services necessary for the design and construction of the Project as set forth in the Design-Build Agreement. Subcontractor, through itself, and Sub-Subcontractors, agrees to provide all construction and other aspects of the Work consistent with the Contract Documents. Design-Builder and Subcontractor agree that to the extent applicable to the performance of the Work hereunder, Subcontractor shall have the same rights, responsibilities, and obligations as to Design-Builder as Design-Builder by the Design-Build Agreement has against and to Owner, except as may be modified herein.

1.3 **Entire Agreement.**

1.3.1 The Contract Documents, all of which are incorporated by reference into this Agreement, form the entire agreement between Design-Builder and Subcontractor and are as fully binding on the parties as if repeated herein. No oral representations or other agreements have been made by the parties except as specifically stated in the Contract Documents.

Article II. Subcontractor's Services and Responsibilities

2.1 **The Work.**

2.1.1 Subcontractor shall furnish, at its own cost and expense, all labor, materials, skills, equipment, scaffolding, power, water, taxes, fees, supplies, tools, facilities, supervision and services to perform the work described in the Contract Documents for each Project (the "Work") except as specifically indicated in the Contract Documents to be the responsibility of others. Subcontractor shall perform the Work in a good and workmanlike manner, free of defects, in compliance with the Contract Documents and all applicable local, state and federal ordinances, laws, rules and regulations, including but not limited to building codes, safety laws and the Occupational Safety and Health Act, and all manufacturers' installation instructions. Without limiting the generality of the foregoing, all construction work to be performed by Subcontractor shall meet or exceed the standards of the industry for commercial construction in the geographic area involved and shall be performed within the time frame as set forth in the Project Schedule.

2.1.2 The Construction Documents shall be deemed to include all revisions thereto as of the date of this Agreement and it shall be Subcontractor's responsibility to stay informed regarding changes in the Construction Documents. Items of work or materials omitted from the Construction Documents that are inferable from the information presented and which are called for by building codes or standard local building practice shall be provided and performed by Subcontractor and are included in the Contract Price and shall be deemed as part of the Work. The description of work to be performed and materials to be furnished by Subcontractor by reference to a section or sections of the plans and specifications referred to in the Construction Documents shall not be deemed to limit the obligations of Subcontractor to perform only such work or furnish only such materials described in such section(s), if work or materials coming within the general description of such section(s) is required by another section of the plans and specifications or from other Contract Documents. Subcontractor shall assume all expenses for engineering services required for the Work.

2.1.3 Subcontractor represents and warrants that prior to commencing a portion of the Work it will have thoroughly examined the Contract Documents and the Site and has ascertained the job site conditions to be encountered in the performance of its obligations including verifications of lines, grades and measurements. Subcontractor represents and warrants that it will enter into this Agreement solely in reliance upon its own information and investigations and not upon any statement or representation made by Design-Builder and further confirms that Subcontractor has all necessary power and authority to enter into and perform all of its obligations under this Agreement. Subcontractor has inspected or shall inspect a Site and the immediate area of Subcontractor's Work prior to submitting bid proposal and before proceeding with the Work, and shall immediately notify Design-Builder in writing of any unacceptable conditions Subcontractor finds. Subcontractor shall verify that all work, storage and access areas and surfaces related to the Work are satisfactory. The commencement of the Work by Subcontractor shall be deemed Subcontractor's acceptance of a Site and all access and storage areas. All services to be performed by Subcontractor shall be completed in the manner and with the quality prevailing among contractors of superior knowledge and skill.

2.1.4 Subcontractor shall also be responsible for verification of locations of all existing utilities and for prevention of damage to the utilities, as well as to follow all applicable encroachment standards affecting the utility rights of way and to adequately protect its own employees and those of other trade contractors and Design-Builder in performing Work.

2.1.5 Subcontractor shall notify Design-Builder in writing of any discrepancy, error, conflict or omission discovered by Subcontractor in the Construction Documents or the work of others, and continuation of the Work subsequent to such discovery shall be at Subcontractor's risk and expense.

2.1.6 All materials to be furnished by Subcontractor shall be new and the best of their respective kinds, except such materials as may be expressly required to the contrary in writing by Design-Builder. Materials and installations shall be in strict accordance with manufacturers' recommendations and industry standards.

2.2 Work of Others.

2.2.1 Subcontractor agrees to reasonably cooperate with, and coordinate its activities so as not to interfere with, those parties performing work the Site, including Owner's and Design-Builder's separate contractors, so that the Project can be completed in an orderly and coordinated manner without unreasonable disruption.

2.2.2 If any part of the Work depends upon other work performed by Design-Builder, or Design-Builder's or Owner's separate contractors, Subcontractor shall, prior to proceeding with that part of the Work, inspect such other work and promptly notify Design-Builder of any discrepancies or defects that would render it unacceptable for Subcontractor's proper performance of the Work. Subcontractor shall not proceed with such part of the Work without further direction from Design-Builder. Design-Builder shall promptly correct or cause to be corrected any such discrepancy or defect in the other work. Subcontractor shall be liable for any losses or damages incurred by an Owner or Design-Builder due to any discrepancies or defects in such other work not reported in writing to Design-Builder by Subcontractor. Commencement of any Work to be performed by Subcontractor constitutes an agreement and affirmation by Subcontractor that the work which preceded Subcontractor's Work has been done in a proper fashion and manner and if any incorrect work by others preceding performance by Subcontractor necessitates all or a portion of Subcontractor's Work to be revised or replaced, it shall be done by Subcontractor at its expense, without any increase in Contract Price.

2.3 Sub-Subcontractors.

2.3.1 Subcontractor shall employ only Sub-Subcontractors who are duly licensed and qualified to perform the Work consistent with the Contract Documents. Subcontractor agrees that each Sub-Subcontractor shall be fully bound to Subcontractor in the same manner as Subcontractor is bound to Design-Builder for all the requirements of the Contract Documents to the extent applicable to the Sub-Subcontractor's scope of work.

2.3.2 Before performance of the Work, Subcontractor shall provide written notice informing Design-Builder the names and addresses of any Sub-Subcontractors working at the Project (including any materialmen and/or suppliers). Subcontractor assumes responsibility to Design-Builder for the proper performance of the Work of Sub-Subcontractors and any acts and omissions in connection with such performance. Subcontractor shall coordinate the activities of all Sub-Subcontractors. Nothing in this Agreement is intended or deemed to relieve Subcontractor from responsibility for the work performed by its Sub-Subcontractors, or create any legal or contractual relationship between Owner or Design-Builder and any Sub-Subcontractor, including but not limited to any third-party beneficiary rights.

2.4 Site Cleanup.

2.4.1 Subcontractor shall perform cleanup of their work area daily and in a diligent manner. Subcontractor shall ensure all contract-specific trash and extra materials are organized or promptly removed and shall immediately remove all lunch or miscellaneous debris and trash. Should Subcontractor fail in any respect to maintain a clean, safe, and orderly work area, Subcontractor shall receive a twenty-four (24) hour written notice. If, upon expiration of the 24 hours, Subcontractor has not complied with the notice to clean, the work shall be performed by others and costs paid from monies due the Subcontractor. Upon Substantial Completion of the Work, or a portion of the Work, Subcontractor shall remove all debris, trash, construction wastes, materials, equipment, machinery and tools arising from the Work or applicable portions thereof to permit Owner to occupy the Project or a portion of the Project for its intended use. Subcontractor shall further contribute four (4) hours of labor time/wk to general jobsite clean-up.

2.5 Inspection.

2.5.1 At all reasonable times, Subcontractor shall provide sufficient facilities for inspection of the Work by Design-Builder at the Site and at all locations where portions of the Work are in progress or various stages of completion. When appropriate portions of the Work are ready for inspection, Subcontractor shall notify Design-Builder in writing.

2.6 Patents and Copyrights.

2.6.1 Subcontractor shall pay all license fees and royalties due for items, materials, methods, systems or processes applicable to the Work.

2.7 Governmental Approvals and Permits.

2.7.1 Subcontractor shall obtain and pay for the necessary permits, approvals, licenses, government charges and inspection fees required for the prosecution of the Work.

2.8 Project Safety.

2.8.1 Subcontractor recognizes the importance of performing the Work in a safe manner so as to prevent damage, injury or loss to (i) all individuals at the Site, whether working or visiting, (ii) the Work, including materials and equipment incorporated into the Work or stored on-Site or off-Site, (iii) the work of others on the Project, and (iv) all other property at the Site or adjacent thereto. Subcontractor assumes responsibility for implementing and monitoring all safety precautions and programs related to the performance of the Work.

2.8.2 Subcontractor and Sub-Subcontractors shall comply with all Legal Requirements relating to safety, as well as any Owner-specific and/or Design-Builder-specific safety requirements set forth in the Contract Documents or established for the Project, provided that such Owner-specific and/or Design-Builder-specific requirements do not violate any applicable Legal Requirement. Subcontractor will immediately report in writing any safety-related injury, loss, damage or accident arising from the Work to Design-Builder and, to the extent mandated by Legal Requirements, to all government or quasi-government authorities having jurisdiction over safety-related matters involving the Project or the Work.

2.9 Warranty.

2.9.1 Subcontractor warrants to Design-Builder that the construction, including all materials and equipment furnished as part of the construction, shall be new unless otherwise specified in the Contract Documents, of good quality, in conformance with the Contract Documents and free of defects in materials and workmanship. Nothing in this warranty is intended to limit any manufacturer's warranty which provides Owner and/or Design-Builder with greater warranty rights than set forth in this Section 2.9 or the Contract Documents. Subcontractor will provide and, if requested, assign to Design-Builder all manufacturers' warranties upon Substantial Completion.

2.10 Correction of Defective Work.

2.10.1 Subcontractor agrees to correct any of the Work that is determined by Design-Builder, in its sole discretion, not to be in conformance with the Contract Documents within a period of two years from the date of Substantial Completion of the Project, or within such longer period to the extent required by any specific warranty included in the Contract Documents or by applicable law.

2.10.2 Subcontractor shall, within forty-eighty (48) hours of receipt of written notice from Design-Builder that the Work is not in conformance with the Contract Documents, take meaningful steps to commence correction of such nonconforming Work, including the correction, removal or replacement of the nonconforming Work and any damage caused to other parts of the Work or the Project affected by the nonconforming Work. If Subcontractor fails to commence the necessary steps within such forty-eight (48) hour period, Design-Builder, in addition to any other remedies provided under the Contract Documents, may provide Subcontractor with written notice that Design-Builder will commence correction of such nonconforming Work with its own forces. If Design-Builder does perform such corrective Work, Subcontractor shall be responsible for all reasonable costs incurred by Design-Builder in performing such correction. If the nonconforming Work creates an emergency requiring an immediate response, the forty-eight (48) period identified herein shall be deemed inapplicable.

2.10.3 The two year period referenced in Section 2.10.1 above applies only to Subcontractor's obligation to correct nonconforming Work and is not intended to constitute a period of limitations for any other rights or remedies Design-Builder may have regarding Subcontractor's obligations under the Contract Documents.

2.10.4 The obligations of Subcontractor under this Section 2.10 shall survive expiration or termination of this Agreement.

2.11 Hazardous Conditions.

2.11.1 Subcontractor is responsible for Hazardous Conditions introduced to the Site by itself, Sub-Subcontractors or anyone for whose acts they may be liable. Subcontractor shall indemnify, defend and hold harmless Owner, Design-Builder and their officers, directors, employees and agents from and against all claims, losses, damages, liabilities, and expenses, including attorneys'

fees and expenses, arising out of or resulting from those Hazardous Conditions introduced to the Site by Subcontractor, Sub-Subcontractors or anyone for whose acts they may be liable.

Article III. Design-Builder's Responsibilities

3.1 Design-Builder shall review submittals, including shop drawings, product data and samples, submitted by Subcontractor. Design-Builder's review of submittals shall be only for the purpose of confirming general conformance with the Construction Documents. Design-Builder's review shall not relieve Subcontractor of its responsibilities to perform the Work in accordance with the Construction Documents.

3.2 Design-Builder shall notify Subcontractor of any errors, inconsistencies, or omissions Design-Builder discovers in the Work. Subcontractor is responsible for any errors, inconsistencies, or omissions in the Work.

Article IV. Ownership of Work Product.

4.1 Work Product.

4.1.1 The Subcontractor shall have no ownership and property rights in any drawings, specifications, and other documents and electronic data ("Work Product") furnished by Design-Builder to Subcontractor under this Agreement. Design-Builder shall be granted ownership of all Work Product, if any, furnished by Subcontractor to Design-Builder under this Agreement.

4.1.2 If either Design-Builder or Subcontractor uses the Work Product furnished to them by the other on any other project, it agrees that it shall do so at its sole risk and without liability or legal exposure to the other party, Owner, or anyone working through them. Such party further agrees that it shall defend, indemnify and hold harmless the other party from and against any and all claims, damages, liabilities, losses and expenses, including attorneys' fees, arising out of or resulting from such use of the Work Product on another project.

Article V. Time of Performance

5.1 Date of Commencement.

5.1.1 The Work shall commence immediately upon Subcontractor's receipt of this Agreement signed by the Design-Builder ("Date of Commencement") unless the parties mutually agree otherwise in writing.

5.2 Time of Completion.

5.2.1 Subcontractor shall diligently and continuously prosecute and complete the Work in accordance with the Project Schedule as it may be revised and issued from time to time during the performance of the Work, and any other scheduling requirements listed in the Contract Documents.

5.2.2 Subcontractor shall participate and cooperate in the development of schedules and other efforts to achieve timely completion of the Work. Subcontractor shall provide Design-Builder information for the scheduling of the times and sequence of operations required for the Work to meet Design-Builder's overall schedule requirements, shall continuously monitor the Project Schedule, including any revisions thereto, so as to be fully familiar with the timing, phasing and sequence of operation of the Work and of other work on the Project, and shall execute the Work in accordance with the requirements of the Project Schedule including any revisions thereto.

5.2.3 Subcontractor shall timely perform the various stages of the Work so that Design-Builder can achieve the dates set forth in the Project Schedule, including any revisions thereto.

5.3 Delays to the Work.

5.3.1 If Subcontractor is delayed in the performance of the Work due to acts, omissions, conditions, events, or circumstances caused by the Design-Builder the time for performance shall be reasonably extended by Change Order *provided that* Subcontractor provides Design-Builder with written notice of Subcontractor's delay within three days following Subcontractor first being made aware of Design-Builder's acts, omissions, conditions, events, or circumstances causing Subcontractor's delay.

5.3.2 Notwithstanding any other provision to the contrary, any delay and resulting damages that arise out of, or relate to, problems caused by Owner or for which Owner is responsible shall be resolved pursuant to Section 12 hereof.

5.3.3 If the Project is delayed due to the Subcontractor or anyone for whom Subcontractor is responsible, and not due to Design-Builder or Owner, Subcontractor shall compensate and indemnify Design-Builder for all costs, damages, and expenses arising from such delay, including but not limited to any liquidated damages or other damages that Owner may assess against Design-Builder which are attributable to Subcontractor or anyone for whom Subcontractor is responsible. In addition, Subcontractor shall, at the direction of Design-Builder and at Subcontractor's own cost and expense, work such overtime and take such other measures as may be necessary to make up for all time lost in the completion of the Project due to such delay.

Article VI. Contract Price

6.1 Contract Price.

6.1.1 Design-Builder shall pay Subcontractor in accordance with Article VII hereof the sum of Twenty One Thousand Four Dollars and Seventy Five cents (\$21,004.75) (the "Contract Price"), subject to adjustments made in accordance with the Contract Documents. Unless otherwise provided in the Contract Documents, the Contract Price is deemed to include all sales, use, consumer and other taxes mandated by applicable Legal Requirements. Design-Builder is not responsible for Subcontractor's bidding or estimating mistakes or miscalculation of market conditions.

6.2 Markups for Changes.

6.2.1 If the Contract Price requires an adjustment due to changes in the Work, the following markups shall be allowed on such changes:

For additive Change Orders, including additive Change Orders arising from both additive and deductive items, it is agreed that Subcontractor mark-ups (aggregating all mark-ups by Sub-Subcontractors of every tier) must not exceed 10% of the value of the Work to be performed by the Subcontractor in the Change Order.

6.2.2 For deductive Change Orders, including deductive Change Orders arising from both additive and deductive items, the deductive amounts shall include Subcontractor's mark-ups.

Article VII. Payment

7.1 Progress Payments.

7.1.1 As set forth in Exhibit D attached hereto, beginning with the first month after the Date of Commencement, Subcontractor shall submit, in accordance with the submission deadline specified in Exhibit D, for Design-Builder's review and approval, Subcontractor's Application for Payment requesting payment for all unpaid and completed Work performed up to the date of submission. The Application for Payment shall be accompanied by all supporting documentation required by the Contract Documents or as may be reasonable requested by the Design-Builder. Failure to submit Subcontractor's Application for Payment by the billing period set forth in this Section 7.1.1, will result in receiving payment the following pay period.

7.1.2 The Application for Payment may request payment for equipment and materials not yet incorporated into the Project, provided that Subcontractor receives Design-Builder's prior written consent and (i) such payment is allowed pursuant to the terms and conditions of the Design-Build Agreement, (ii) Design-Builder is satisfied that the equipment and materials are suitably stored at either the Site or another acceptable location, (iii) the equipment and materials are protected by suitable insurance, and (iv) upon payment, Design-Builder will receive the equipment and materials free and clear of all liens and encumbrances.

7.1.3 The Application for Payment shall constitute Subcontractor's representation that the Work has been performed consistent with the Contract Documents, has progressed to the point indicated in the Application for Payment, and that title to all Work will pass to Owner free and clear of all claims, liens, encumbrances, and security interests upon the incorporation of the Work into the Project, or upon Subcontractor's receipt of payment, whichever occurs earlier.

7.1.4 Design-Builder shall make payment on Subcontractor's properly submitted and accurate Application for Payment within Seven (7) days after Design-Builder's receipt of the funds from the Owner less the total of payments previously made, and less amounts properly withheld under the Contract Documents or Arizona law. Design-Builder's receipt of the funds from the Owner is a condition precedent to Design-Builder's obligation to pay Subcontractor pursuant to the terms of this Agreement or other Contract Document. Design-Builder and Subcontractor agree that Design-Builder will pay Subcontractor directly from the funds Design-Builder receives from the Owner and Design-Builder is not obligated to pay Subcontractor from Design-Builder's own funds.

7.2 Retainage on Progress Payments.

7.2.1 Unless otherwise specifically stated in an exhibit to this Agreement, Design-Builder will retain Ten percent (10%) from each of Subcontractor's Application for Payment. Retainage will be included in Design-Builder's final payment to Subcontractor, conditioned upon Design-Builder having received such retained amounts from an Owner. Design-Builder's receipt of the retained amount from the Owner is a condition precedent to Design-Builder's obligation to pay Subcontractor pursuant to the terms of this Agreement or other Contract Document.

7.3 Withholding of Payments.

7.3.1 If Design-Builder determines that Subcontractor is not entitled to all or part of an Application for Payment, it will notify Subcontractor in writing. The notice shall indicate the specific amounts Design-Builder intends to withhold, the reasons and contractual basis for the withholding, and the specific measures Subcontractor must take to rectify Design-Builder's concerns. Design-Builder and Subcontractor will attempt to resolve Design-Builder's concerns. If the parties cannot resolve such concerns, Design-Builder shall pay Subcontractor the uncontested amount of the Application for Payment, and Subcontractor may pursue its rights under the Contract Documents.

7.4 Final Payment.

7.4.1 Subcontractor shall submit its Final Application for Payment to Design-Builder in accordance with Section 7.4.2 below. Design-Builder shall make payment on Subcontractor's properly submitted and accurate Final Application for Payment within Seven (7) days after Design-Builder's receipt of final payment from Owner, provided also that Subcontractor has satisfied the requirements for final payment set forth in Section 7.4.2 below.

7.4.2 At the time of submission of its Final Application for Payment, Subcontractor shall provide the following information:

7.4.2.1.1 If requested by Design-Builder, an affidavit that there are no claims, obligations or liens outstanding or unsatisfied for labor, services, material, equipment, taxes or other items performed, furnished or incurred for or in connection with the Work which will in any way affect Design-Builder's or Owner's interests;

7.4.2.1.2 A conditional release upon final lien waiver, in a form conforming to the statutes, rules and regulations of the state in which the Project is located and approved by Design-Builder, executed by Subcontractor waiving, upon receipt of final payment by Design-Builder, all claims, except those claims previously made in writing to Design-Builder and remaining unsettled at the time of final payment;

7.4.2.1.3 All operating manuals, warranties and other deliverables required by the Contract Documents; and

7.4.2.1.4 Upon request by Design-Builder, certificates of insurance confirming that required coverages will remain in effect consistent with the requirements of the Contract Documents.

7.5 Advance Payments.

7.5.1 Design-Builder has the right, at its sole option, to advance any payment due Subcontractor under this Agreement.

7.6 Payment Not Acceptance.

7.6.1 No payment to Subcontractor under this Agreement shall be evidence of, or construed to be, acceptance of defective, faulty, improper or non-conforming work.

7.7 Subcontractor's Payment Obligations.

7.7.1 Subcontractor will pay its Sub-Subcontractors the amounts Subcontractor has received from Design-Builder on account of their work. Subcontractor will impose similar requirements on its Sub-Subcontractors to pay those parties with whom they have contracted. Subcontractor will indemnify and defend Owner and Design-Builder against any claims for payment and mechanic's liens arising from or related to any Sub-Subcontractor's work on a Project.

7.8 Record Keeping and Finance Controls.

7.8.1 During the performance of the Work and for a period of three (3) years after final payment of the Work, Design-Builder and Design-Builder's accountants shall be afforded access to and the right to audit from time-to-time, upon reasonable notice, Subcontractor's records, books, correspondence, receipts, subcontracts, purchase orders, vouchers, memoranda and other data relating

to the changes in the Work, all of which Subcontractor shall preserve for a period of three (3) years after final payment. Any multipliers or markups agreed to by Subcontractor and Design-Builder as part of this Agreement are only subject to audit to confirm that such multiplier or markup has been charged in accordance with this Agreement.

Article VIII. Stop Work and Termination

8.1 Design-Builder's Right to Stop Work.

8.1.1 Design-Builder may, without cause and for its convenience, order Subcontractor in writing to stop and suspend the Work.

8.1.2 Subcontractor is entitled to seek an adjustment of the Contract Price and/or times for completion of the Work if its cost or time to perform the Work has been adversely impacted by any suspension or stoppage of work by Design-Builder. Notwithstanding anything to the contrary herein, if Design-Builder's suspension of the Work is the result of Owner's suspension of Design-Builder's work under the Design-Build Agreement, then Design-Builder shall pay Subcontractor only those amounts Design-Builder actually receives from Owner on account of the Work.

8.2 Design-Builder's Right to Terminate for Convenience.

8.2.1 Upon ten (10) days' written notice to Subcontractor, Design-Builder may, for its convenience and without cause, elect to terminate this Agreement. In such event, Design-Builder shall have the right to use the existing Work Product, if any, for purposes of completing the Project, and shall pay Subcontractor for all Work executed.

8.2.2 If Design-Builder's termination of Subcontractor for convenience is the result of Owner's termination of Design-Builder for convenience under the Design-Build Agreement, then Design-Builder shall pay Subcontractor only those amounts Design-Builder actually receives from Owner on behalf of Subcontractor.

8.3 Design-Builder's Right to Terminate for Cause.

8.3.1 If Subcontractor fails to (i) provide a sufficient number of skilled workers, (ii) supply the materials required by the Contract Documents, (iii) comply with applicable Legal Requirements, (iv) timely pay, without cause, its Sub-Subcontractors, (v) prosecute the Work with promptness and diligence to ensure that the Work is completed in accordance with the Project Schedule, as such schedule may be adjusted, or (vi) perform material obligations under the Contract Documents, then Design-Builder shall have the rights, in addition to any other rights and remedies provided in the Contract Documents or by law, set forth in Sections 8.3.2 and 8.3.3 below.

8.3.2 Upon the occurrence of an event set forth in Section 8.3.1 above, Design-Builder may provide written notice to Subcontractor that it intends to terminate the Agreement unless the problem cited is cured, or commenced to be cured, within Twenty-Four (24) hours of Subcontractor's receipt of such notice. If Subcontractor fails to cure, or reasonably commence to cure, such problem, then Design-Builder may declare this Agreement terminated for default by providing written notice to Subcontractor of such declaration.

8.3.3 Upon declaring this Agreement terminated pursuant to Section 8.3.2 above, Design-Builder may enter upon the premises and take possession, for the purpose of completing the Work, of all materials, equipment, scaffolds, tools, appliances and other items thereon, which have been purchased or provided for the performance of the Work, all of which Subcontractor hereby transfers, assigns and sets over to Design-Builder for such purpose, and to employ any person or persons to complete the Work and provide all of the required labor, services, materials, equipment and other items. In the event of such termination, Subcontractor shall not be entitled to receive any further payments under the Contract Documents until the Work shall be finally completed in accordance with the Contract Documents. At such time, if the unpaid balance of the Contract Price exceeds the cost and expense incurred by Design-Builder in completing the Work, such excess shall be paid by Design-Builder to Subcontractor. If Design-Builder's cost and expense of completing the Work exceeds the unpaid balance of the Contract Price, then Subcontractor shall be obligated to pay the difference to Design-Builder. Such costs and expense shall include not only the cost of completing the Work, but also losses, damages, costs and expenses, including attorneys' fees and expenses, incurred by Design-Builder in connection with the procurement and defense of claims arising from Subcontractor's default.

8.3.4 If Design-Builder improperly terminates the Agreement for cause, the termination for cause will be converted to a termination for convenience in accordance with the provisions of Section 8.2 of this Agreement.

8.4 Subcontractor's Right to Stop Work.

8.4.1 If Owner fails to pay amounts due Design-Builder under the Design-Build Agreement for Work performed by Subcontractor, such failure is not due to the fault of Subcontractor, and Subcontractor has not been paid such amounts due Subcontractor stop work in accordance with Section 8.4.2.

8.4.2 Subcontractor shall provide Design-Builder with written notice that Subcontractor will stop work unless said failure to pay the amount is cured within seven (7) days from Design-Builder's receipt of Subcontractor's notice. If Design-Builder does not cure the problem within such seven (7) day period, Subcontractor may stop work. To the extent Design-Builder's failure to pay is related to a dispute between the parties, the dispute will be resolved and the parties will continue performance in accordance with Article 12.

8.5 Subcontractor's Right to Terminate for Cause.

8.5.1 Subcontractor, in addition to any other rights and remedies afforded under the Contract Documents or at law, may terminate the Agreement for cause in accordance with Section 8.5.2 below if Design-Builder fails to cure the problems set forth in Section 8.4.1 above within thirty (30) days after Subcontractor has stopped the Work.

8.5.2 Upon the occurrence of the event set forth in Section 8.5.1 above, Subcontractor may provide written notice to Design-Builder that it intends to terminate the Agreement unless the problem cited is cured, or commenced to be cured, within seven (7) days of Design-Builder's receipt of such notice. If Design-Builder fails to cure, or reasonably commence to cure, such problem, then Subcontractor may give a second written notice to Design-Builder of its intent to terminate within an additional seven (7) day period. If Design-Builder, within such second seven (7) day period, fails to cure, or reasonably commence to cure, such problem, then Subcontractor may declare this Agreement terminated for default by providing written notice to Design-Builder of such declaration. In such case, Subcontractor shall be entitled to recover in the same manner as if Design-Builder had terminated this Agreement for its convenience under Section 8.2 above.

Article IX. Insurance and Bonds

9.1 Subcontractor's Insurance Requirements:

9.1.1 Subcontractor is responsible for procuring and maintaining, from insurance companies authorized to do business in the state in which the Project is located, the insurance coverages set forth in the Insurance Exhibit D attached to this Agreement, with the minimum ratings set forth in said Exhibit, for certain claims which may arise from or out of the performance of the Work and obligations under the Contract Documents.

9.1.2 Subcontractor shall require its Sub-Subcontractors to procure and maintain, from insurance companies authorized to do business in the state in which the Project is located, the insurance coverages set forth in the Insurance Exhibit attached hereto as Exhibit D..

9.1.3 Subcontractor's and its Sub-Subcontractors' insurance coverage set forth in the Insurance Exhibit shall specifically delete any design-build or similar exclusions that could compromise coverages because of the design-build delivery of the Project.

9.1.4 Prior to commencing any services under this Agreement, Subcontractor shall provide Design-Builder with certificates evidencing that (i) all insurance obligations required by the Contract Documents are in full force and in effect and will remain in effect for the duration required by the Contract Documents and (ii) no insurance coverage will be canceled, renewal refused, or materially changed unless at least thirty (30) days prior written notice is given to Design-Builder.

9.1.5 Except as otherwise stated in the Insurance Exhibit, the insurance policies required herein shall list Design-Builder and Owner, and any other entities required by the Contract Documents, if any, as an additional insured.

9.1.6 If any of the foregoing coverages are required to remain in force after final payment, an additional certificate evidencing continuation of such coverage shall be submitted with the final Application for Payment.

9.2 Waiver of Subrogation.

9.2.1 Design-Builder and Subcontractor waive against each other and Owner, Sub-Subcontractors, Owner's or Design-Builder's separate contractors, agents and employees of each and all of them, all damages covered by property insurance provided herein, except such rights as they may have to the proceeds of such insurance. Design-Builder and Subcontractor shall, where appropriate, require similar waivers of subrogation from Sub-Subcontractors and separate contractors of Design-Builder, and shall require each of them to include similar waivers in their contracts. These waivers of subrogation shall not contain any restriction or limitation that will impair the full and complete extent of its applicability to any person or entity unless agreed to in writing.

9.3 Bonds and Other Performance Security.

9.3.1 Subcontractor shall provide the performance bonds and labor and material payment bonds or other performance security as specifically set forth in an exhibit to this Agreement.

Article X. Indemnification

10.1 Patent and Copyright Infringement.

10.1.1 Subcontractor shall defend any action or proceeding brought against Owner or Design-Builder based on any claim that the Work, or any part thereof, or the operation or use of the Work or any part thereof, constitutes infringement of any United States patent or copyright, now or hereafter issued. Design-Builder shall give prompt written notice to Subcontractor of any such action or proceeding and will reasonably provide authority, information and assistance in the defense of same. Subcontractor shall indemnify and hold harmless Owner and Design-Builder from and against all damages and costs, including but not limited to attorneys' fees and expenses awarded against Owner or Design-Builder in any such action or proceeding. Subcontractor agrees to keep Design-Builder informed of all developments in the defense of such actions.

10.1.2 If Owner is enjoined from the operation or use of the Work, or any part thereof, as the result of any patent or copyright suit, claim, or proceeding, Subcontractor shall at its sole expense take reasonable steps to procure the right to operate or use the Work. If Subcontractor cannot so procure such right within a reasonable time, Subcontractor shall promptly, at Subcontractor's option and at Subcontractor's expense, (i) modify the Work so as to avoid infringement of any patents, or copyrights, or (ii) replace said Work with Work that does not infringe or violate any such patent or copyright.

10.1.3 Sections 10.1.1 and 10.1.2 above shall not be applicable to any suit, claim or proceeding based on infringement or violation of a patent or copyright (i) relating to a particular process or product of a particular manufacturer specified by Owner or Design-Builder or (ii) arising from modifications to the Work by Owner or Design-Builder after acceptance of the Work. If the suit, claim or proceeding is based upon events set forth in the preceding sentence, Design-Builder shall defend, indemnify and hold harmless Subcontractor to the same extent Subcontractor is obligated to defend, indemnify and hold harmless Design-Builder in Section 10.1.1 above.

10.1.4 The obligations set forth in this Section 10.1 shall constitute the sole agreement between the parties relating to liability for infringement or violation of any patent or copyright.

10.2 Payment Claim Indemnification.

10.2.1 Subcontractor shall indemnify, defend and hold harmless Owner and Design-Builder from any claims or mechanic's liens brought against Owner, Design-Builder, or against the Project as a result of the failure of Subcontractor, or those for whose acts it is responsible, to pay for any services, materials, labor, equipment, taxes or other items or obligations furnished or incurred for, or in connection with the Work. Within Twenty-Four (24) hours of receiving written notice from Design-Builder that such a claim or mechanic's lien has been filed, Subcontractor shall commence to take the steps necessary to discharge said claim or lien and shall diligently pursue to completion, including, if necessary, the furnishing of a mechanic's lien bond. If Subcontractor fails to do so, Design-Builder will have the right to discharge the claim or lien and hold Subcontractor liable for costs and expenses incurred, including attorneys' fees.

10.3 Subcontractor's General Indemnification.

10.3.1 Subcontractor shall indemnify, defend and save harmless Design-Builder and Owner from and against any and all claims, debts, demands, damages (including direct, liquidated, consequential, incidental or other damages), judgments, awards, losses, liabilities, interest, attorney's fees, costs and expenses of whatsoever kind or nature at any time arising out of any failure of Subcontractor to perform any of the terms and conditions of this Agreement or which are in any manner directly or indirectly caused or occasioned by or contributed to, or claimed to be caused or occasioned by, or contributed to, by any act, omission, fault or negligence, whether active or passive, of Subcontractor or anyone acting under its direction or control, or on its behalf in connection with or incident to the work performed by Subcontractor or anyone acting under its direction or control; even though the same may have resulted from the joint, concurring or contributory act, omission or negligence whether active or passive, of Design-Builder, Owner or any other person, except Subcontractor is not obligated to indemnify, defend, or hold harmless Design-Builder for any claim caused by the negligence or fault, the breach or violation of a statute, ordinance, governmental regulation, standard, or rule, or the breach of contract of Design-Builder, its agent or employee, or any third party under the control or supervision of Design-Builder. Without limiting the generality of the foregoing, the same shall include injury or death to any person or persons, including agents and employees of Owner, Design-Builder and Subcontractor, and damage to any property, regardless of location, including property of Owner, Design-Builder and Subcontractor, and shall extend to any similar obligations of Design-Builder undertaken by it under the

Design-Build Agreement with respect to the work hereunder. Subcontractor will, on request, and at its expense, defend any action, suit or proceedings arising hereunder and shall reimburse and pay Design-Builder for any loss, cost, damage or expense (including legal fees) suffered by it.

Article XI. Changes to the Contract Price and Time

11.1 Owner Generated Changes.

11.1.1 If Owner issues changes affecting the Work, Subcontractor agrees, if directed by Design-Builder, to meet with Design-Builder and Owner to review and discuss the changes. Subcontractor shall only be entitled to adjustments in its Contract Price and the times for completion of the Work attributable to such Owner-generated changes to the extent Design-Builder actually receives such adjustments from Owner. If Subcontractor disputes the adjustment, such dispute shall be resolved pursuant to Article 12 of this Agreement.

11.2 Design-Builder Generated Changes.

11.2.1 Changes to the scope of work for the Project issued by Design-Builder shall be governed by the provisions set forth in the following sections of this Article 11.

11.3 Change Orders.

11.3.1 A Change Order is a written instrument issued after execution of the Agreement signed by Design-Builder and Subcontractor, stating their agreement upon all of the following:

11.3.1.1 The scope of the change in the work;

11.3.1.2 The amount of the adjustment to the Contract Price; and

11.3.1.3 The extent of the adjustment to the times for completion of the work;

11.3.2 All changes in the Work authorized by applicable Change Order shall be performed under the applicable conditions of the Contract Documents. Design-Builder and Subcontractor shall negotiate, in good faith and as expeditiously as possible, the appropriate adjustments for such changes.

11.4 Work Change Directives.

11.4.1 A Work Change Directive is a written order prepared and signed by Design-Builder directing a change in the work prior to agreement on an adjustment in the Contract Price and/or the times for completion of the work.

11.4.2 Design-Builder and Subcontractor shall negotiate, in good faith and as expeditiously as possible, the appropriate adjustments for the Work Change Directive. Upon reaching an agreement, the parties shall prepare and execute an appropriate Change Order reflecting the terms of the agreement.

11.5 Minor Changes in the Work.

11.5.1 Minor changes in the Work are changes that do not involve an adjustment in the Contract Price and/or times for completion of the Work and do not materially and adversely affect the Work, including the design, quality, performance and workmanship required by the Contract Documents. Design-Builder may make minor changes in the Work consistent with the intent of the Contract Documents, provided, however, that Design-Builder shall promptly inform Subcontractor, in writing, of any such changes.

11.6 Contract Price Adjustment.

11.6.1 The increase or decrease in Contract Price resulting from a change in the Work shall be determined by one or more of the following methods:

11.6.1.1 Unit prices set forth in this Agreement or as subsequently agreed between the parties;

11.6.1.2 A mutually accepted, lump sum, properly itemized and supported by sufficient substantiating data to permit evaluation by Design-Builder;

11.6.1.3 Costs, fees and any other markups set forth in this Agreement; and

11.6.1.4 If an increase or decrease cannot be agreed to as set forth in items 11.6.1.1 through 11.6.1.3 above and Design-Builder issues a Work Change Directive, the cost of the change of the Work shall be determined by the reasonable expense and savings in the performance of the Work resulting from the change, including a reasonable overhead and profit, as may be set forth in this Agreement. If the net result of both additions and deletions to the Work is an increase in the Contract Price, reasonable overhead and profit shall be calculated on the basis of the net increase to the Contract Price. Subcontractor shall maintain a documented, itemized accounting evidencing the expenses and savings associated with such changes.

11.6.2 If Design-Builder and Subcontractor disagree upon whether Subcontractor is entitled to be paid for any services required by Design-Builder, or if there are any other disagreements over the scope of Work or proposed changes to the Work, Design-Builder and Subcontractor shall resolve the disagreement pursuant to Article 12 hereof. As part of the negotiation process, Subcontractor shall furnish Design-Builder with a good faith estimate of the costs to perform the disputed services in accordance with Design-Builder's interpretations. If the parties are unable to agree and Design-Builder expects Subcontractor to perform the services in accordance with Design-Builder's interpretations, Subcontractor shall proceed to perform the disputed services, conditioned upon Design-Builder issuing a written order to Subcontractor (i) directing Subcontractor to proceed and (ii) specifying Design-Builder's interpretation of the services that are to be performed.

11.7 Emergencies.

11.7.1 In any emergency affecting the safety of persons and/or property, Subcontractor shall act, at its discretion, to prevent threatened damage, injury or loss. Any change in the Contract Price and/or times for completion of the Work on account of emergency work that cannot be agreed to between the parties shall be determined as provided in Article 12.

Article XII. Contract Adjustments and Disputes

12.1 Requests for Contract Adjustments and Relief.

12.1.1 If either Subcontractor or Design-Builder believes that it is entitled to relief against the other for any event arising out of or related to the Work or a Project, such party shall provide written notice to the other party of the basis for its claim for relief. Subcontractor shall provide Design-Builder written notice of claims for which Owner may be responsible in sufficient time for Design-Builder to meet its notice requirements to Owner set forth in the Design-Build Agreement. In the absence of any specific notice requirement, written notice shall be given within a reasonable time, not to exceed ten (10) days, after the occurrence giving rise to the claim for relief or after the claiming party reasonably should have recognized the event or condition giving rise to the request, whichever is later. Such notice shall be in accordance with the Contract Documents and shall include sufficient information to advise the other party of the circumstances giving rise to the claim for relief, the specific contractual adjustment or relief requested and the basis of such request.

12.2 Dispute Avoidance and Resolution.

12.2.1 The parties are fully committed to working with each other and agree to communicate regularly with each other at all times so as to avoid or minimize disputes or disagreements. If disputes or disagreements do arise, Subcontractor and Design-Builder each commit to resolving such disputes or disagreements in an amicable, professional and expeditious manner so as to avoid unnecessary losses, delays and disruptions to the Work.

12.3 Disputes Involving Owner.

12.3.1 To the extent a claim, dispute or controversy arises out of, or relates to, problems caused by Owner or for which Owner is responsible ("Owner Disputes"), such Owner Disputes shall be resolved pursuant to the dispute resolution clause set forth in the Design-Build Agreement. Both Design-Builder and Subcontractor agree to cooperate in the presentation and prosecution or defense of Owner Disputes.

12.4 Disputes Not Involving the Owner.

12.4.1 For any claim, dispute or controversy not arising out of, or relating to, problems caused by Owner or for which Owner is responsible, Subcontractor and Design-Builder will first attempt to resolve such claim, dispute or controversy at the field level through discussions between Design-Builder and Subcontractor.

12.4.2 If the claim, dispute or controversy cannot be resolved on terms satisfactory to both parties, the parties shall submit the claim, dispute or controversy to non-binding mediation. The mediation shall be conducted by a mutually agreeable impartial mediator, or if the parties cannot so agree, a mediator designated by the American Arbitration Association ("AAA") pursuant to its Construction Industry Mediation Rules. The mediation will be governed by and conducted pursuant to a mediation agreement negotiated by the parties or, if the parties cannot so agree, by procedures established by the mediator. Unless otherwise mutually agreed by Design-Builder and Subcontractor and consistent with the mediator's schedule, the mediation shall commence within ninety (90) days of the submission of the dispute for mediation. Persons with authority to resolve the dispute shall be present at the mediation.

12.5 Arbitration.

12.5.1 Any claims, disputes or controversies between the parties arising out of or relating to this Agreement, or the breach thereof, which have not been resolved in accordance with the procedures set forth in Section 12.4 above, shall be decided by arbitration in accordance with the Construction Industry Arbitration Rules of the AAA then in effect, unless the parties mutually agree otherwise.

12.5.2 The award of the arbitrator(s) shall be final and binding upon the parties without the right of appeal to the courts. Judgment may be entered upon it in accordance with applicable law by any court having jurisdiction thereof.

12.5.3 Subcontractor and Design-Builder expressly agree that any arbitration pursuant to this Section 12.5 may be joined or consolidated with any arbitration involving any other person or entity (i) necessary to resolve the claim, dispute or controversy, or (ii) substantially involved in or affected by such claim, dispute or controversy. Both Design-Builder and Subcontractor will include appropriate provisions in all contracts they execute with other parties in connection with the Project to require such joinder or consolidation.

12.5.4 The prevailing party in any arbitration, or any other final, binding dispute proceeding upon which the parties may agree, shall be entitled to recover from the other party reasonable attorneys' fees and expenses incurred by the prevailing party.

12.6 Duty to Continue Performance.

12.6.1 Unless provided to the contrary in the Contract Documents, Subcontractor shall continue to perform the Work and Design-Builder shall continue to satisfy its payment obligations to Subcontractor, pending the final resolution of any dispute or disagreement between Design-Builder and Subcontractor.

Article XIII. Miscellaneous

13.1 Assignment.

13.1.1 Subcontractor shall not, without the written consent of the Design-Builder, assign, transfer or sublet any portion or part of the Work or the obligations required by the Contract Documents.

13.2 Successorship.

13.2.1 Design-Builder and Subcontractor intend that the provisions of the Contract Documents are binding upon the parties, their employees, agents, heirs, successors, and assigns.

13.3 Governing Law.

13.3.1 The Agreement and all Contract Documents shall be governed by the laws of the state in which the Project is located.

13.4 Severability.

13.4.1 If any provision or any part of a provision of the Contract Documents shall be finally determined to be superseded, invalid, illegal, or otherwise unenforceable pursuant to any applicable Legal Requirements or court order, such determination shall not impair or otherwise affect the validity, legality, or enforceability of the remaining provision or parts of the provision of the Contract Documents, which shall remain in full force and effect as if the unenforceable provision or part were deleted.

13.5 No Waiver.

13.5.1 The failure of either Design-Builder or Subcontractor to insist, in any one or more instances, on the performance of any of the obligations required by the other under the Contract Documents shall not be construed as a waiver or relinquishment of such obligation or right with respect to future performance.

13.6 Headings.

13.6.1 The headings used in this Agreement, or any other Contract Document, are for ease of reference only and shall not in any way be construed to limit or alter the meaning of any provision.

13.7 Notice.

13.7.1 Whenever the Contract Documents require that notice be provided to the other party, notice will be deemed to have been validly given (i) if delivered in person to the individual intended to receive such notice, (ii) four (4) days after being sent by registered or certified mail, postage prepaid to the address indicated in this Agreement, (iii) if transmitted by facsimile, by the time stated in a machine generated confirmation that notice was received at the number of the intended recipient; or (iv) if delivered by email, and the recipient acknowledges having received the email.

13.8 Amendments.

13.8.1 The Contract Documents may not be changed, altered, or amended in any way except in writing signed by a duly authorized representative of each party.

13.9 Survival.

13.9.1 Subcontractor's obligations under this Agreement shall not be released and shall specifically survive the completion of all services hereunder by Subcontractor, final payment to Subcontractor, and the termination of this Agreement for any reason.

Article XIV. Electronic Data

14.1 Electronic Data.

14.1.1 The parties recognize that Contract Documents, including drawings, specifications and other Work Product may be transmitted among Design-Builder, Subcontractor and others in electronic media as an alternative to paper hard copies (collectively "Electronic Data").

14.2 Transmission of Electronic Data.

14.2.1 Design-Builder shall determine, after consultation with Subcontractor, the software and the format for the transmission of Electronic Data. Each party shall be responsible for securing the legal rights to access the agreed-upon format, including, if necessary, obtaining appropriately licensed copies of the applicable software or electronic program to display, interpret and/or generate the Electronic Data.

14.2.2 Neither party makes any representations or warranties to the other with respect to the functionality of the software or computer program associated with the electronic transmission of Work Product. Unless specifically set forth elsewhere in this Agreement, ownership of the Electronic Data does not include ownership of the software or computer program with which it is associated, transmitted, generated or interpreted.

14.2.3 By transmitting Work Product in electronic form, the transmitting party does not transfer or assign its rights in the Work Product. The rights in the Electronic Data shall be as set forth in Article 4 of the Agreement. Under no circumstances shall the transfer of ownership of Electronic Data be deemed to be a sale by the transmitting party of tangible goods.

14.3 Electronic Data Protocol.

14.3.1 The parties acknowledge that Electronic Data may be altered or corrupted, intentionally or otherwise, due to occurrences beyond their reasonable control or knowledge, including but not limited to compatibility issues with user software, manipulation by the recipient, errors in transcription or transmission, machine error, environmental factors, and operator error. Consequently, the parties understand that there is some level of increased risk in the use of Electronic Data for the communication of design and construction information and, in consideration of this, agree, and shall require their independent contractors, Subcontractors and to agree, to the following protocols, terms and conditions set forth in this Section 14.3.

14.3.2 Electronic Data will be transmitted in the format determined in Section 14.2.1 above, including file conventions and document properties, unless prior arrangements are made in advance in writing.

14.3.3 The Electronic Data represents the information at a particular point in time and is subject to change. Therefore, the parties shall agree upon protocols for notification by the author to the recipient of any changes which may thereafter be made to the Electronic Data, which protocol shall also address the duty, if any, to update such information.

14.3.4 The transmitting party specifically disclaims all warranties, expressed or implied, including, but not limited to, implied warranties of merchantability and fitness for a particular purpose, with respect to the media transmitting the Electronic Data. However, transmission of the Electronic Data via electronic means shall not invalidate or negate any duties pursuant to the applicable standard of care with respect to the creation of the Electronic Data, unless such data is materially changed or altered after it is transmitted to the receiving party, and the transmitting party did not participate in such change or alteration.

14.4 In the event the Design-Build Agreement contains a provision governing Electronic Data, and there is a conflict between the provision in the Design-Build Agreement and this Article 14, the provision in the Design-Build Agreement takes precedence, notwithstanding the order of precedence set forth in Section 1.4.2.

Article XV. Confidential Information

15.1 Confidential and/or Proprietary Information.

15.1.1 Confidential Information is defined as information which is determined by the transmitting party to be of a confidential or proprietary nature and: (i) the transmitting party identifies as either confidential or proprietary; (ii) the transmitting party takes steps to maintain the confidential or proprietary nature of the information; and (iii) the document is not otherwise available in or considered to be in the public domain. The receiving party agrees to maintain the confidentiality of the Confidential Information and agrees to use the Confidential Information solely in connection with a Project.

15.1.2 Subcontractor may receive information from Design-Builder that is either confidential or proprietary to either Design-Builder or to Owner. Such information shall be labeled as confidential and/or proprietary. Subcontractor agrees to maintain the confidential nature of such information and to execute any such additional agreements as may be required by Owner or Design-Builder with respect to such information.

15.1.3 In the event the Design-Build Agreement contains a provision governing Confidential Information, and there is a conflict between the provision in the Design-Build Agreement and this Article 16, the provision in the Design-Build Agreement takes precedence.

Article XVI. Other Provisions

16.1 Other provisions, if any, are as follows:

The Exhibits attached to this Agreement, which include, but are not limited to the following:

- Exhibit A – Subcontractor’s Scope of Work;*
- Exhibit B – Preliminary Lien Information;*
- Exhibit C – Plan Coversheet & Plan Page List;*
- Exhibit D – Insurance & Required Documents.*
- Exhibit E – Geotechnical Report*

Document Name	Document Link	File Type	Document Comment
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Cost Code	Company	Contact	Phone	Email	Amount
31-25000	Desert Services	Jared Aiken	989-330-7859	jared@desertservices.net	\$21,004.75

EXHIBIT A

SCOPE OF WORK: SWPPP

In the event that there exists a conflict in the contract documents, then the highest quality and the greatest quantity shall prevail. Subcontractor shall warrant the Work to be free from all defects or deviations for a period of a MINIMUM of (2) year from Date of Substantial Completion or as stipulated within the contract documents, whichever is longer.

1. Provide, all **SWPPP** turnkey complete, as further defined below and AHJ code requirements.
2. Sales tax is excluded.
3. A payment and performance bond is not required and is not included.
4. Overhead and profit for scope changes shall not exceed 10% as per Owner Contract.
5. Retainage of 10% will be withheld as outlined in the subcontract agreement.

Payment Application

A detailed Pay Application Schedule of Values (SOV) must be submitted within (7) days of receipt of this work order.

Applicable Specifications

Subcontractor shall comply with all project specifications applicable and pertinent to Subcontractor's work, but shall pay particular attention to the following:

Division 01 – General Requirements

Definitions

The following definitions shall include all costs for labor, materials, equipment, overhead, profit, trade layout, insurance, services, supplies, fuel, scaffolding, ladders, staging, tools, small tools, hoisting of any kind, lull, forklifts, boom lifts, temporary protection, pumps, safety apparatus, rigging, and any other facilities necessary (unless specifically excluded further below) in accordance with the Contract Documents (which is this Agreement, the plans, and specifications):

1. The word "**Furnish**" means to purchase and deliver to a location on the project site designated by Contractor's Superintendent, but not to install, because it will be installed by others.
2. The words "**Install Only**" or "**Installed Only**" means to take custody of material furnished by others from a location on the project site designated by Contractor's Superintendent and install it completely including final connections as required throughout.
3. The word "**Provide**" means to purchase and deliver material/equipment to the project site and then install that material/equipment completely including final connections as required throughout.

Project Requirements

1. **Provide** SWPPP narrative design manual per EPA specifications.
2. **Provide** ADEQ approved SWPPP sign and spill kit.
3. **Provide** filter fabric for storm drain inlet protection.
4. **Provide** silt fence or an EPA approved equal per plan.
5. **Provide** stabilized construction entrance with rock and filter fabric per plan.
6. **Provide** concrete roll off system.
7. **Provide** dust control sign.



8. **Provide** and maintain SWPPP book per ADEQ specifications.
 - a. Perform visual inspection of the construction site once every fourteen (14) days and log/maintain findings on the site conditions of the SWPPP controls and note and correct any erosion problem areas. **21 inspections included**
 - b. Perform visual inspection within 24 hours of rainfall greater than or equal to one-half inch (1/2"). Maintain and log findings on the site conditions of the SWPPP controls and note and correct any erosion problem areas as required.

General Requirements

1. Subcontractor shall include mobilizations as required to maintain or better the schedule as requested by LGE Project Superintendent.
2. Subcontractor shall provide all labor and equipment necessary to complete their scope of work in such a time as to not delay the General Contractor's schedule attached to this agreement. The subcontractor has agreed to provide and maintain the manpower necessary so as not to delay this project in any way. No additional monies will be paid for overtime though this project may require that overtime be worked to maintain the construction schedule. Subcontractor is solely responsible for maintaining their production and installation schedules to meet the Project's Construction Schedule per your scope of work. Delays by other subcontractors or contractor may constitute overtime due to their delay and contractor may pay for premium time for lost time if required to catch the schedule back up.
3. Subcontractor shall monitor, coordinate, and order material such as to avoid material price increases. Subcontractor shall notify Contractor of impending material price increases with sufficient time to allow for pre-purchasing, ordering ahead, stockpiling on or off site, or other avenues to avoid cost increases. "Sufficient time" shall be defined as a minimum of fourteen (14) calendar days for the purpose of this requirement.
4. Subcontractor shall be independently, solely, and fully responsible in all aspects for the QA/QC of their work. Subcontractor shall employ an individual responsible for ensuring work is completed in a workmanlike fashion and to industry standards. Subcontractor shall, upon completion of an area of work or specific work items, complete, sign, and submit to Contractor a QA/QC form, signifying that the work has been checked, is in compliance with the contract documents, and meets industry standards for workmanship.
5. Subcontractor shall treat the exposed building slab and walls as a finished product. All markings must be removed, and no permanent types of mediums will be allowed, including but not limited to, spray paint, permanent markers, red chalk, etc. It is this Subcontractor's responsibility to take all necessary precautions to fully protect the building slab from damage, both structural and aesthetic in nature. Subcontractor shall not, under any circumstances, apply adhesive tape of any type to a polished or finished concrete surface. No exceptions.
6. Subcontractor shall ensure that any equipment brought onto the building slab is equipped with non-marking tires or wheel cover "booties" and a system to capture and contain any leaking fluids.
7. Subcontractor shall not be permitted/allowed vehicle parking of any type on the building slab. Special circumstances or requirements require a written request submitted to the LGE Project Manager for consideration. This request shall be submitted a minimum of seven (7) calendar days ahead of the anticipated need. All violating subcontractors shall share costs for any oil or stain clean-up.
8. Subcontractor shall include all necessary precautions to protect adjacent surfaces (panels, windows, canopies, sidewalks, paving, etc.) and prevent damage of existing work or products in place while performing this scope of work is included. Subcontractor shall be liable for repair costs due to damage while performing this scope of work.
9. Subcontractor shall provide protection for this scope of work until work is accepted by Owner.
10. Subcontractor shall mark shut-offs, breakers, or other methods of electrical, water, or gas disconnect that are located above/behind acoustical ceilings or access panels with adhesive stickers. The adhesive stickers shall be color-coded green for water, blue for gas, and red for electrical. The stickers shall be applied to the nearest adjacent surface.



11. Subcontractor shall furnish and install access panels, as necessary for access to Subcontractors' system and as appropriate for the partition or structure where the panel is being installed.
12. Subcontractor shall accept, offload, and inventory, any materials purchased by others for use by this subcontractor. Any shortages or damage must be immediately reported to the Contractor in writing. Shortages or damage not immediately reported shall become the responsibility of this subcontractor.
13. Subcontractor shall furnish a minimum of 2% attic stock, or one (1) full unopened container of each color, specification, or design. All attic stock will be furnished in new, unopened, and clearly identified containers, unless partial containers are required to meet quantity requirements. All attic stock shall be placed in a location TBD by the LGE Superintendent. A transmittal shall be provided with delivery of the materials, to be signed by the LGE site representative. A copy of this signed transmittal shall be submitted electronically to the Contractor. Below are examples of finished products to be provided. This list is not complete and if a Subcontractor is unsure of the materials or quantities, it is the Subcontractor's responsibility to clarify.
 - a. Paint, wallcovering, lamps, tile, laminate, carpet (all flooring in general), vinyl base, acoustical tile, stone/faux stone, anything with a dye lot.
14. Subcontractor shall be responsible for sealing any penetrations, suitable to the partition, wall, or structure being penetrated.
15. Subcontractor confirms they have reviewed and will comply with all UL Requirements, UL Assembly Report, Fire Rating Requirements, Acoustical Requirements, and Wall Assemblies per the contract documents.
16. Subcontractor shall coordinate this scope of work with ALL drawing sets and specifications, including but not limited to the architectural, general, structural, civil, plumbing, mechanical, electrical, landscape sheets and invitation to bid instructions. Subcontractor shall coordinate with all other trades, systems, building components, vendors, LGE project management team, and Project Superintendent prior to the ordering and installation of work to avoid and prevent conflicts.

If routing / hanging conflicts occur, Subcontractor shall provide additional labor, equipment and materials as required to reroute around conflict. Change requests for coordination issues will not be accepted.
17. Subcontractor acknowledges the scope of work for this project is based upon the contract documents. While the contract documents do not necessarily fully describe or indicate specifics regarding the scope of work identified in the "Requirements" section of this agreement, your firm has reviewed the drawings to compile a complete scope of work outside of the specific exclusions noted in the "Exclusions" section of this agreement. Anything inferred or implied in the drawings or as necessary to complete this scope of work has been included.
18. Subcontractor shall verify dimensions, clearances, and locations of all work with sufficient time to comply with the construction schedule.
19. Subcontractor shall provide all engineering, layout, field dimensions and/or staking for this scope. Benchmarks, building corners, and property lines will be staked by the General Contractor.
20. Subcontractor shall provide delivery and unloading of own material. Coordinate unload/storage locations, and delivery schedule with LGE Design Build Superintendent. Subcontractor shall be responsible for securing all materials prior to installation and all equipment left onsite.
21. Subcontractor shall keep city streets and access roads clean from dirt and track out due to own vehicles, equipment, and material delivery trucks. Subcontractor shall either hand sweep or mechanically sweep the streets to avoid conflicts with the City or County in the event of track out.
22. Subcontractor shall provide all necessary dust control measures for this scope of work, as defined by the AHJ. Driving onsite will be limited to 5 MPH, or as designated by the LGE Project Superintendent.
23. Subcontractor shall provide blue staking for own scope of work. All subcontractors shall be singularly responsible for own blue stake notifications and coordination.
24. Subcontractor shall schedule all city inspections and inform LGE Project Superintendent. When scheduling an inspection, a 24-hour notice is required for all Engineering and material testing needs. Testing and cost of passing testing by other. Retesting/reinspection costs shall be the responsibility of the Subcontractor.
25. Subcontractor shall ensure equipment, materials and work areas are organized, clean, tidy, and safe at all times. Subcontractors are responsible for cleanup on the job site. If the subcontractor has NOT cleaned up daily then a back charge for the full cost of labor to complete the clean-up work may be charged via a deductive Commitment Change Order.



26. Subcontractor shall submit "RFI" Request for Information pertaining to any and all discrepancies and conflicts noted in the construction documents. Any scope of work installed without receiving clear direction from the design team will be removed and reinstalled by this subcontractor. Delays due to untimely RFI submission will not be acceptable and will be the responsibility of the Subcontractor to recover the time.
27. Subcontractor shall coordinate and verify existing work in place is suitable for the start of this scope of work. Notify LGE Project Superintendent in writing of deficient scope of work preventing the on time start of work. By proceeding with installation, the subcontractor accepts existing conditions.
28. Subcontractor shall provide temporary generator power and task lighting as needed to complete this scope of work.
29. Subcontractor shall log into and thoroughly review all project related construction documents regularly to ensure construction is being installed with the most current information. The following documents include but are not limited to, drawings, specifications, RFI's, ASI's, PR's, SK's, submittals, schedules located in LGE's "Procore Technologies Inc." document control database. Contact LGE's project team should you not have access to view these listed documents. Email notifications shall be provided of any changes to subcontractors.
30. Subcontractor shall ensure jobsite supervisors or foreman are in attendance at all field project coordination, scheduling, quality control and safety meetings, when requested. Attendees shall have the authority to make scheduling, sequencing, coordination, quality, safety, and any other project related decisions. Additionally, a representative from the Subcontractor's company shall be present at all inspections of the subcontractor's work.
31. Subcontractor shall be fully responsible to ensure:
 - a. Record drawings are kept current and updated weekly at a minimum. Final as-built drawings are to be provided upon completion of the scope of work.
 - b. All submittals, shop drawings, samples and mockups are a requirement of this contract and are reviewed as a consideration to all parties. No part of this process shall be considered or understood to transfer any part of the responsibility for full compliance with the contract documents from or away from the Subcontractor. Do not procure or manufacture materials and products until submittals, shop drawings, samples and mockups have been reviewed and returned unless specifically directed by LGE Project Manager.
 - c. All labor warranties, manufacturer warranties, and O&M manuals shall be turned in not later than thirty (30) days after return of reviewed submittals, unless otherwise agreed upon and memorialized in the Specific Scope Inclusions. All warranties shall state the beginning of the warranty period is "upon substantial completion".
 - d. Subcontractor shall perform a warranty punch list after the project completion. In the event any warranty punch list items require repair or replacement due to defective workmanship and/or materials, Subcontractor shall repair or replace said items and the (2) year correction period for those items repaired and/or replaced shall be renewed and recommenced.
32. Subcontractor shall understand and agree that drawings noting "By Contractor" or "By GC", or "By General Contractor" shall be by this subcontractor if relating to this trade.
33. Subcontractor shall follow LGE project Superintendent's directions regarding parking, access, staging, material delivery and jobsite rules. If materials are delivered to the jobsite prior to the scheduled installation date, Subcontractor may be required to provide labor and equipment to relocate materials on site if it is impeding the work of others at no additional costs to LGE. Subcontractor shall provide labor and equipment to relocate materials within 24 hours of LGE Project Superintendent's request, in the event that Subcontractor does not complete the request within the timeframe specified LGE reserves the right to relocate the material on the Subcontractor's behalf with no liability for damage/loss and expenses may be charged via a deductive Commitment Change Order.
34. Subcontractor shall conduct all operations during normal business hours, as defined by the LGE Project Superintendent and/or AHJ. If required, any noisy and invasive scope of work by this Subcontractor will be provided outside normal business hours. Scheduled shut down and tie-in with utilities, which require off hours or overtime by this subcontractor are provided and included in this Subcontract.



Changes in Scope of Work / Alternates

1. Subcontractor shall not deviate from the construction documents or perform any change in scope of work until or unless receiving written direction from the LGE Project Manager.
2. Change requests will follow Prime Contract requirements. Subcontractor shall submit all change requests with back-up to fully and completely substantiate the requested change. Change request shall be submitted within seven (7) calendar days of notification, or within seven (7) calendar days of Subcontractor discovering a requirement for additional work. Changes submitted without this back up, or outside of the specified timeframe, shall be returned and not considered valid change request. This back up shall include, but not be limited to:
 - a. Copy of the direction for change, I.E. RFI, delta plan change, LGE written direction, etc.
 - b. Copy of the affected area of the original plan, pre change.
 - c. Copy of the affected area of the plan, showing the change.
 - d. Full breakdown of labor, material, equipment, and OH&P.
3. Subcontractor shall understand and agree that T&M work is defined as follows; all resources required to complete the directed task shall be accounted for and billed on an actual cost basis (cost, time, equipment, materials, etc.) plus the contractually allowable mark up.
 - a. Subcontractor shall be singularly and solely responsible to ensure tickets are kept on a daily basis, documenting all labor, equipment, and materials used. Tickets shall be submitted daily for approval and signature by the on-site LGE representative; tickets shall not be backdated, nor submitted for approval more than 24 hours after the fact. Upon completion of the T&M work, signed tickets shall be submitted within seven (7) calendar days of the completion of the work with a change order request to the LGE Project Manager for issuance of a contract change order.
4. Without exception, Subcontractor's change requests shall clearly indicate and show separately actual costs and contractually allowable mark-up.

Alternates

1. Credit to not install stabilized construction entrance: (\$2,475.00)/EA
2. Wire backed silt fence (if absolutely required): \$4.90/LF
3. Credit to not provide dust control sign: (\$575.00)/EA
4. Concrete Roll Off: \$725.00

Schedule

Subcontractor will begin work when notified by LGE Design Build project team and shall coordinate and execute work in accordance with construction documents and project management team's written direction. A baseline schedule is provided with this subcontract agreement for duration scheduling, sequence and milestone information for scope of work. LGE's weekly updated short interval schedule will take precedent over all other schedules. This schedule will be available in LGE's "Procore Technologies Inc." document control database. In the event you do not have access, please contact LGE's project team in writing to obtain access. Scope durations below:

Safety

Provide personal protective equipment at all times. This shall include OSHA compliant eye protection, face masks, hard hats, reflective shirts or vest, long pants, shirt with sleeves, and footwear. Subcontractors shall conduct work in a safe and professional manner consistent with OSHA rules and regulations. No radios or headphones will be permitted on this jobsite. Subcontractor will provide a site-specific safety plan and activity hazard analysis for this scope of work.



Shop Drawings, Mockups, & Submittals

Provide shop drawings, mockups, and submittals within 4 weeks after contract is issued. Do not procure or manufacture materials and products until reviewed submittals, shop drawings, samples, and mockups have been approved. All submittals, shop drawings, samples, and mockups shall be in full compliance with the construction documents. Notify LGE project team in writing if additional time is needed to procure submittal information. No exceptions will be considered without a formal letter of substitution request. Any substitution request shall be submitted on a separate coversheet and include any cost impacts.

Exclusions

1. Performance and payment bonds
2. Sales tax



Pre-lien Information:

JOB: International Furniture Direct

JOB SITE: 7839 North 73rd Avenue
Glendale, Arizona 85303

GENERAL
CONTRACTOR: LGE Corporation dba LGE Design Build
1200 North 52ND Street
Phoenix, Arizona 85008
PHONE: 480-966-4001

OWNER: Chicago International Real Estate LLC
D/b/a International Furniture Direct
5750 McEver Rd Ste. A
Oakwood, GA 30566

LENDER: Commercial Banking
8181 Communications Parkways Bldg b Floor 5
Plano, TX 75024-0239

ESTIMATED COMPLETION DATE: TBD

NO PERFORMANCE BOND REQUIRED

GLENDAL, AZ 85303

20240816 - INTERNATIONAL FURNITURE OBJECT (SKELETON), AZEES COGNOM - A000 - COVER SHEET Printed Date: 9/15/2025 11:11 AM



EXHIBIT D – INSURANCE REQUIREMENTS & REQUIRED DOCUMENTS

Hello LGE Subcontractor,

The following insurance documents must be received by our Accounting Department at insurance@lgedesignbuild.com within 10 days from the date of your signed contract, or this contract is voidable.

A certificate of insurance naming LGE Corporation dba LGE Design Build as additional insured with respects to the following:

- a. **Commercial Liability:**
 - Each Occurrence - \$1,000,000
 - Personal & Adv Injury - \$1,000,000
 - General Aggregate - \$2,000,000
 - Products – Comp/OP AGG - \$2,000,000
- b. **Automobile Liability:**
 - Combined Single Limit each accident - \$1,000,000
- c. **Umbrella Liability:**
 - Each Occurrence - \$2,000,000
 - Aggregate - \$2,000,000
- d. **Workers Compensation:**
 - E.L. Each Accident – 1,000,000
 - E.L. Each Disease – Each Employee - \$1,000,000
 - E.L. Disease – Policy Limit - \$1,000,000
- e. **Contractors Pollution:**
 - Each Occurrence - \$1,000,000
 - Aggregate - \$1,000,000
- f. **Professional Liability (if applicable):**
 - Each Claim - \$1,000,000
 - Aggregate - \$1,000,000
- g. Subcontractor shall procure insurance to meet the requirements as set forth in Owner's Contract. ** Insurance requirements are subject to change. **
- h. For COI renewals please send to: insurance@lgedesignbuild.com

Please reference the Project Directory in **Procore** for all scheduling, shop drawings, submittals and R F I ' s

Furthermore, please submit:

Dust Permit and all written Hazardous Communication Programs.

- Project Specific S.D.S. sheets.
- Written Project Specific Safety Program.
- Maricopa County air quality subcontractor registration number.

If your company digs, paves, or moves dirt in any way, include a copy of your onsite personnel's Dust Compliance training ID card to Jim Kelly at jimk@lgedesignbuild.com within five (5) days of receipt of your contract or prior to commencement of your portion of the work for this project, whichever is earliest.

BILLINGS SHALL BE SUBMITTED IN PREMIER TO LGE NO LATER THAN THE 22ND OF EACH MONTH. LATE BILLINGS CANNOT BE ACCEPTED. PLEASE CONTACT YOUR ASSIGNED PROJECT COORDINATOR PRIOR TO THE 22ND FOR ANY ASSISTANCE NEEDED.

When submitting a billing, please upload a signed pay application along with your signed conditional progress/final waiver in the attachment field. Banks no longer accept typed signatures. All signatures must be hand signed, or a digitally created e-signature backed by certificate in software such as Adobe or Bluebeam.

If any offsite stored material is being billed, then the owner (and bank if there is a loan) will need the following submitted in the subcontractor's billing please:

- **INVOICES:** Material invoice(s) from the supplier (NOT SUBCONTRACTOR) with cost matching what the subcontractor pay app is submitted for. Invoices must be clear identifying the stored material billed for this project. There should be no markup added in the payment application for the material invoices.
- **INSURANCE:** Certificate of insurance with the owner and bank if there is a loan for this project listed as the additional insured. The insurance cert must specify the stored material is insured for at least the cost of what is billed.
- **PHOTOS:** Reasonable photo evidence of all specific material that is stored for this project only and billed for. Any offsite stored bulk materials not specifically labeled and separated for this project will likely not be accepted by the bank/owner for payment.

LOCATION: Business name, contact, address, phone number of company/location where the material is stored.

It is important that the LGE billing procedures are followed to ensure prompt payment.

If the LGE Construction Manager requests extra work, you must submit a proposal for the work. An approved Work Order will then be issued by the LGE Construction Department, and a copy will be emailed to your office. All billings for extra work will be submitted on your invoice with a copy of the "Approved" work order, and only when the work is 100% complete. All billings and waivers for base contract and approved extra work shall be submitted through Premier.

PLEASE DO NOT MAIL INVOICES IF YOU HAVE ALREADY SUBMITTED THEM.

We ask that you please make sure that our Accounting Department has your correct email address. You will receive monthly memos notifying you of all jobs to be "Finalized", as well as "First Draws" for new projects and/or any other pertinent information.

Should you have any questions or need clarification regarding the above-mentioned billing procedures, please contact Mike Scott at our office or by email at mikes@lgedesignbuild.com. Phone extension #2202

For Payment information please contact Stephanie Alvarez at stephaniea@lgedesignbuild.com. Phone extension #2212

Sincerely – LGE Team



EXHIBIT D
INSURANCE REQUIREMENTS & REQUIRED DOCUMENTS

Hello LGE Subcontractor,

The following insurance documents must be received by our Accounting Department at insurance@lgedesignbuild.com within 10 days from the date of your signed contract, or this contract is voidable.

A certificate of insurance naming LGE Corporation dba LGE Design Build as additional insured with respects to the following:

- a. **Commercial Liability:**
 - Each Occurrence - \$1,000,000
 - Personal & Adv Injury - \$1,000,000
 - General Aggregate - \$2,000,000
 - Products – Comp/OP AGG - \$2,000,000
- b. **Automobile Liability:**
 - Combined Single Limit each accident - \$1,000,000
- c. **Umbrella Liability:**
 - Each Occurrence - \$2,000,000
 - Aggregate - \$2,000,000
- d. **Workers Compensation:**
 - E.L. Each Accident – 1,000,000
 - E.L. Each Disease – Each Employee - \$1,000,000
 - E.L. Disease – Policy Limit - \$1,000,000
- e. **Contractors Pollution:**
 - Each Occurrence - \$1,000,000
 - Aggregate - \$1,000,000
- f. Professional Liability (if applicable):
 - Each Claim - \$1,000,000
 - Aggregate - \$1,000,000
- g. Subcontractor shall procure insurance to meet the requirements as set forth in Owner's Contract. ** Insurance requirements are subject to change. **
- h. For COI renewals please send to: insurance@lgedesignbuild.com

Please reference the Project Directory in **Procore** for all scheduling, project coordination, change orders, shop drawings, submittals and any other items having to do with the construction of this project.

Furthermore, please submit:

- Written Project Specific Safety Program.

Invoices SHALL BE EMAILED TO gc@lgedesignbuild.com NO LATER THAN 30 days after performance of work. Please reference the LGE Project number on your invoice. Late billings will not be accepted.

It is important that the LGE billing procedures are followed to ensure prompt payment.

If the LGE Construction Manager requests extra work, you must submit a proposal for the work. An approved Work Order will then be issued by the LGE Construction Department, and a copy will be emailed to your office. All billings for extra work will be submitted on your invoice with a copy of the "Approved" work order, and only when the work is 100% complete.

We ask that you please make sure that our Accounting Department has your correct email address.

For Payment information please contact Matt Erickson at matthewe@lgedesignbuild.com

Sincerely – LGE Team

Exhibit E

REPORT ON GEOTECHNICAL INVESTIGATION

DESIGNATION: International Furniture Direct

LOCATION: SEC 73rd Avenue & Northern Avenue
Glendale, Arizona

CLIENT: LGE Design Build

PROJECT NO: 242091SA

DATE: December 2, 2024





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APPENDIX – Field and Laboratory Data



1.0 INTRODUCTION

This report presents the results of a subsoil investigation carried out at the site of the proposed *International Furniture Direct* development, to be constructed at the southeast corner of 73rd Avenue and Northern Avenue in Glendale, Arizona. The general location of the proposed development is shown in Figure 1.0.1.

We understand that construction will consist of an 86,272 square foot warehouse building on 6.49 acres of undeveloped farmland. The building will be single story (32' clear height) with slab on grade, tilt panel walls and dock-high truck bays on the south side of the building. Structural loads are expected to be light to moderate and no special considerations regarding settlement tolerances are known at this time. Adjacent areas will be landscaped or paved to support light to moderate passenger and truck traffic. Landscaped areas will be utilized for storm water retention and disposal.



Figure 1.0.1 General Site Location

2.0 GENERAL SITE AND SOIL CONDITIONS

2.1 Site Conditions

The subject site is currently undeveloped farmland. The property is bounded to the north by Northern Avenue, to the south by a parking lot, to the east by a commercial building, and to the west by 73rd Avenue. At the time of the investigation, the site contained soil berms (approximately 1.5 to 2 feet high) that run north to south, approximately every 85 feet. The soil berms were used for irrigation purposes and to separate crops of alfalfa. A concrete-lined irrigation canal is located along the northern boundary of the site and runs parallel to Northern Avenue. The property appears to be relatively flat and level, and the ground surface elevation of the site appears to be generally consistent with the surrounding areas. Refer to the following site photos for more details.



Figure 2.1.1 Facing North



Figure 2.1.2 Facing South

A cursory review of available historical aerial imagery was conducted for the property. Based on the historical images, the site was used for agricultural purposes since at least 1949. It is difficult to determine from the aeriels, but a single structure appears to be evident along the northern boundary of the site in 1954, and was removed by 1964. By 1976 Northern Avenue was cleared and graded and subsequently expanded and paved by 2001. By 2008 Augusta Avenue and the *Bascom Hunter Technologies, Inc.* building appear to have been developed. Sometime between 2020 and 2022 the commercial building to the east (*Villa Dolce Phoenix*) was developed. By early 2024 the southern adjoining property was developed into a parking lot. Refer to the following historical aerial photos:



Figure 2.1.3 Dated 1949



Figure 2.1.4 Dated 1969

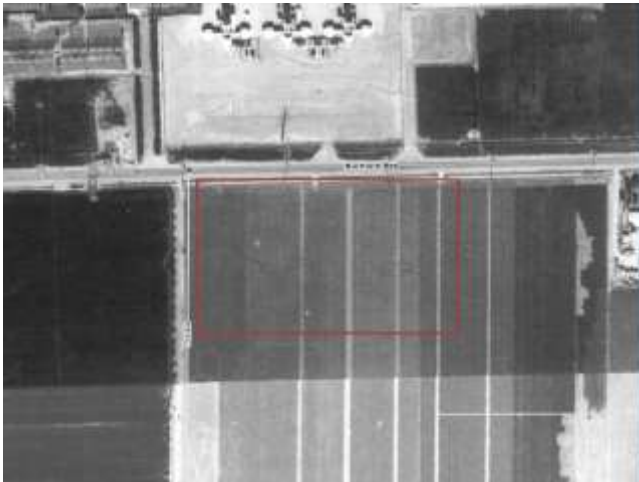


Figure 2.1.5 Dated 1976



Figure 2.1.6 Dated 2001



Figure 2.1.7 Dated 2008



Figure 2.1.8 Dated 2022

“Historical Aerial Photography,” *ArcGIS Web Application*. [Online]. Available: <https://gis.maricopa.gov>.

2.2 Geological Conditions

While the site is not located in an area of reported “considerable” subsidence, it is located within the known area that has undergone minor subsidence due to groundwater removal, specifically the West Valley Land Subsidence area. Areas of subsidence are known under certain conditions to produce ground surface earth fissuring. However, subsidence is a basin-wide phenomenon that typically results in gradual differential elevation changes over long distances, and which would not affect the type of buildings proposed for this site. No evidence of earth fissures was observed on the site. Fissure gullies are typically localized and form over subsurface irregularities such as buried bedrock ridges, causing tensional stresses in the overlying sediments. Where such anomalies are not present, subsidence tends to be uniform over a wide area, thus having minimal effects on surficial structures. The nearest documented earth fissures are around Luke Air Force base, approximately 8 miles west of this site. Based on local experience, earth fissures historically have not been a problem within the subject site.

If any cracks, crevasses, or fissures are noted during site excavation this office should be notified immediately. Representatives from this office will then visit the site, assess the features, and make recommendations regarding restorative measures.

2.3 Seismic Design Parameters

The project area is in a seismic zone that is considered to have low historical seismicity. The seismicity of the Phoenix area has had only two magnitude 3.0 events in over 100 years. Liquefaction is not considered a concern as groundwater exceeds 50 feet below ground surface.

Although borings were not advanced to 100 feet, based on the nature of the subsoils encountered in the borings, seismic surveys from adjacent sites, and geology in the area, Site Class Definition, Class D may be used for design of the structures. In addition, the following seismic parameters may be used for design (based on ASCE 7-16 (IBC2018), utilizing the ATC Hazards by location online tool).

Table 2.3.1 Seismic Parameters

MCE ¹ spectral response acceleration for 0.2 second period, S _s :	0.183g
MCE ¹ spectral response acceleration for 1.0 second period, S ₁ :	0.067g
Site coefficient, F _a :	1.6
Site coefficient, F _v :	2.4
MCE ¹ spectral response acceleration adjusted for site class, S _{MS} :	0.292g
MCE ¹ spectral response acceleration adjusted for site class, S _{M1} :	0.160g
5% Damped spectral response acceleration, S _{DS} :	0.195g
5% Damped spectral response acceleration, S _{D1} :	0.107g
NOTE 1: MCE = maximum considered earthquake	

2.4 General Subsurface Conditions

S&A obtained information about the subsurface conditions at the site using eight 7-inch diameter hollow-stem auger soil borings to record the soil conditions and collect samples for laboratory testing. Borings B-1 through B-6 were placed in the proposed building's footprint and were drilled to the intended advancement depths of approximately 11.5 to 21.5 feet below the existing ground surface (bgs). Borings SG-1 and SG-2 were placed in the proposed new pavement and parking areas and were drilled to 5 feet bgs.

Although not indicated on the logs, fill or disturbed soil is likely present at the subject site due to farming activities on and around the site. Lacking telltale signs of fill it is difficult to distinguish re-worked native soils from in-situ native. The soil profile generally consists of sandy lean clay, sandy silt, clayey sand, and occasionally well-graded gravel with sand to the depth of the maximum depth of exploration at up to 21.5 feet bgs. In addition, subordinate amounts of gravel and trace to weak amounts of calcareous cementation were noted in the soil profile. Standard Penetration Resistance Test (SPT) values were inconsistent and ranged from 8 to 50+ blows per foot (bpf). No groundwater was encountered during this investigation. Based on visual and tactile observation, the upper soils were in a 'dry to moist' to 'moist' state at the time of the investigation.

Laboratory testing indicates in-situ dry densities of the upper soils range from 88.7 to 114.8 pounds per cubic foot (pcf) with water contents on the order of 4.8 to 12.1 percent at the time of investigation. Liquid limits range from 26 to 38 percent with plasticity indices of 10 to 18 percent. The upper soils exhibit a volume increase (**swell**) due to wetting of **2.8 percent** when compacted to moisture and density levels normally expected during construction. 'Undisturbed' samples displayed moderate (1.8 to 2.8 percent) compression under initial incremental loading to a maximum confining load of 3,200 pounds per square foot (psf) and **significant (6.3 to 6.9 percent)** additional compression due to inundation (**hydro-collapse**)

3.0 ANALYSIS AND RECOMMENDATIONS

3.1 Analysis

Analysis of the field and laboratory data indicates that subsoils at the site are suitable for the support of the proposed structures on shallow foundations and slab-on-grade construction subject to some **remedial ground improvement earthworks**. It is assumed that grades for the building pads will remain essentially the same (± 3 feet), apart from any old irrigation ditches, which may need to be filled in.

The site is currently under cultivation. These farming activities usually result in the upper 12 to 18 inches being disturbed by plowing. Laboratory and field testing indicate that the primary geotechnical concern is the presence of soils with low and variable densities that are susceptible to additional compression due to inundation (**hydro-collapse**). In addition, farming activities have disturbed the upper soils. These conditions could cause excessive settlement resulting in damage to overlying structures. Accordingly, recommendations are made to over-excavate and re-compact the bearing soils to increase density and reduce the potential for collapse. The over-excavated and re-compacted soil will mitigate, **but not eliminate** the potential for additional settlement if the deeper soils become wet. It is paramount to provide proper drainage to limit the potential for water infiltrating under slabs. Accordingly, creating and maintaining positive drainage from natural water sources away from the buildings is imperative. This includes during construction when foundations are open and most vulnerable.

The swell potential of the fine portion of the **upper clayey soils is a concern**. The potential is strong enough to cause differential movements of slabs-on-grade such as floors, sidewalks, patios and lightly loaded foundation but not enough to cause damage to heavier structures. Accordingly, it is paramount to provide proper drainage to limit the potential for water infiltrating under slabs. A minimum slope of at least 5 percent for 10 feet is recommended for unpaved landscaped areas.

Typical recommendations to **reduce (not eliminate)** the swell potential include reducing the compaction requirements and requiring higher moisture contents during pad preparation and requiring at least **12-inches of non-expansive material** to be placed directly beneath the conventional building slabs and slabs contiguous to the structure such as sidewalks.

The native subgrade soils are considered suitable for lime stabilization. As an alternate to placing 12 inches of non-expansive **common borrow** import to complete the building pad, the native soils may be treated with chemical lime slurry (a minimum of 8 inches for lower subgrade modulus) to reduce the

plasticity and swell potential to less than 1.5 percent. Commercial lime slurry is recommended to reduce the environmental concerns for blowing lime dust and to ensure proper hydration and mixing. The specified under-slab aggregate base course should not be eliminated. If this option is considered, it is recommended that a mix design be conducted using the native, subgrade soils and varying amounts of proposed lime to determine an optimum amount. For further recommendation on lime stabilization refer to Section 3.9 of this report.

For standard shallow foundations to perform as expected, attention must be paid to provide proper drainage to limit the potential for water infiltration of deeper soils. It is assumed that the landscape plan will use mostly low water use or "green" desert type plants (xeriscape). It is preferred to keep irrigated plants at least 5 feet away from structures with irrigation schedules set and maintained to run intermittently. **Unpaved planter areas should be sloped at least 5 percent for a distance of at least 10 feet away from the building.** It is understood that this may not be possible due to ADA maximum slope requirements for the adjacent sidewalks and patios. The slope may be reduced to 2 percent provided extra care is taken to ensure sidewalks and other hardscape features do not create a "dam" that prevents positive drainage away from the buildings, creating a "pond" adjacent to the building. Roof drainage should also be directed away from the building in paved scuppers. Pre-cast loose splash blocks should not be used as they can be dislodged and/or eroded. Roof drains should not be allowed to discharge into planters adjacent to the structure. It is preferred that they be directed to discharge to pavement (per photo example), retention basins or discharge points located at least 10 feet away from the building.



It is reiterated that shallow spread footings may be used for the exterior walls and other light interior columns since this is the most economical system available. However, this shallow system relies on the dry strength of the unsaturated native soils. A limited depth of re-compaction is recommended to increase density of the near surface soils that are more likely to encounter seasonal moisture changes, or deeper foundations. **The deeper native soils are moisture sensitive and could experience differential settlement if subjected to significant surface water infiltration.** Recognizing the need to minimize significant water penetration adjacent to the building perimeter that could detrimentally impact the building foundation, the following additional recommendations are made to protect foundations:

1. Take extra precaution to backfill and compact native soil fill to 95 percent in all exterior wall locations.
2. Avoid utility trenches passing through retention basins leading to the building. If unavoidable, backfill the trench with MAG Section 728 ½-sack CLSM to cut off preferred drainage paths.
3. Avoid placing retention basins next to building foundations. **A distance of at least 10 feet should be maintained between structures and the location of any retention basin maximum fill level and 15 feet from any UST.**

4. Create and maintain positive drainage away from the exterior wall for a minimum of 10 feet.
5. Avoid sidewalks, curbs or other elements that create a dam that could cause water to pond within 5 feet of the perimeter wall.
6. Include no irrigated landscape materials in the first 3 feet next to the building.
7. Between 3 feet and 5 feet, include only landscape materials that can be irrigated with a maximum of 1 gallon per hour emitter heads. Set and maintain irrigation controllers to prevent 24/7 flows.
8. Any landscape materials requiring greater than 1 gallon per hour irrigation, including turf, shall be at least 5 feet from the outside face of the building.
9. All irrigation feeder lines, other than those that supply individual emitters, shall not be placed closer than 5 feet to the building.

Groundwater in the vicinity is reported to be more than 100 feet deep and is not expected to be a factor in the design or construction of foundations and underground utilities. Excavation operations should be relatively straightforward although deeper excavations may encounter dense and/or cemented soils which may impede progress and possibly require the use of heavier equipment.

For exterior slabs-on-grade, frequent jointing is recommended to control cracking and reduce tripping hazards should differential movement occur. It is also recommended to pin the landing slab to the building floor/stem wall. This will reduce the potential for the exterior slab lifting to block the operation of out-swinging doors. Pinning typically consists of 24-inch-long No. 4 reinforcing steel dowels placed at 12-inch centers.

3.2 Site Preparation

Depending on the time of year of construction and irrigation schedule, the surface soils may be wet and require time to dry and/or stabilization prior to their re-use. **Irrigation operations should cease as far ahead of construction start-up as possible.** If the soils are still wet when construction begins, the exposed grade **may become unstable under excavating equipment.** Additional time may be required to allow the soil to dry when exposed, or alternative methods such as lime stabilization or rock fill may be required to stabilize the exposed grade. Likewise special attention must be paid to backfilling existing irrigation ditches. The soft wet material should be removed, generally to a depth of 1 to 2 feet, re-placed and compacted.

The entire area to be occupied by the proposed construction should be stripped of all vegetation, debris, rubble, obviously loose surface soils and any other deleterious materials. Oversize material such as concrete irrigation lining may be broken and placed in landscape area fills. If plant stalks are present, they should be brushed from the site (i.e. do not turn them into the surface soil). The root systems may be

incorporated into the fill. Due to the disturbance of the surface soils, **at least the upper 12 inches (and possibly more if loose/unstable soils are encountered)** of native soils should be pre-compacted within the entire building footprint prior to placing fills. Depending on equipment used, it may be necessary to partially over-excavate in order to get a full 12-inch compacted surface. Final grades should provide space for the 12 inches of non-expansive material for the top of building pad.

Subsoils should be further over-excavated at least **2 feet** below proposed footing bottom elevation, **or existing grade, whichever is deeper**, extending at least 5 feet beyond the footing edges within all footing areas. The entire building pad does not require deep over-excavation if footing lines can be accurately located during earthwork operations. A representative of the Geotechnical Engineer should examine the subgrade once sub-excavation is complete and prior to backfilling to ensure removal of deleterious materials. Fill placement and quality should be as defined in the "Fill and Backfill" section of this report.

Consideration should also be given to extending the over-excavation out beyond the perimeter of the building and/or pavement areas to support heavy crane loads during building erection. Alternatively, a designated crane path could be developed, and additional earthwork performed to support crane loading and tilt-panel lifting, prior to earthwork equipment leaving the site.

Prior to placing structural fill below footing bottom elevation, the exposed grade should be scarified to a depth of 8 inches, moisture-conditioned to optimum (± 2 percent) and compacted to at least 95 percent of maximum dry density as determined by ASTM D-698. Pavement areas should be scarified, moisture-conditioned and compacted in a similar manner, **increasing the depth of scarification to 12 inches**, to compensate for any loose areas related to the previous farming activities.

All cut areas and areas above footing bottom elevation that are to receive floor slab only fill should be scarified 8 inches, moisture conditioned to at least optimum to 3 percent above optimum and lightly but uniformly compacted to 90 but not more than 95 percent of maximum dry density as determined by ASTM D-698.

3.3 Foundation Design

The following bearing capacities can be utilized for design:

Table 3.3.1 Foundation Bearing Capacities

Option	Foundation Type	Foundation Depth ⁽¹⁾	Bearing Medium	Bearing Capacity	Comments
Minor Structures	Spread	1.5 ft.	Compacted Subgrade	1,500 psf	2
Main Structures	Spread	2.0 ft.	2 ft. Eng. Fill	2,500 psf	3

- Comments:
- 1. Foundation Depth refers to the minimum depth to bottom of footing elevation below existing site elevation or lowest finished grade within 5 feet, whichever is deeper.
 - 2. For minor structures such as screen walls, planter walls, canopies etc. not connected to any main structure. The bottom of footing excavation should moisture-conditioned to optimum (± 2 percent) and compacted to at least 95 percent of maximum dry density as determined by ASTM D-698.
 - 3. Shallow spread footings bearing on at least **2 feet of engineered fill** plus 8 inches of scarification and re-compaction. Continuous and square footings, subject to the full allowable bearing pressure, **should not exceed 7 feet and 15 feet** respectively to stay within settlement tolerances. If any isolated column load exceeds 560 kips, then contact our office for other options. Refer to figure 3.3.1 for spread footing detail.

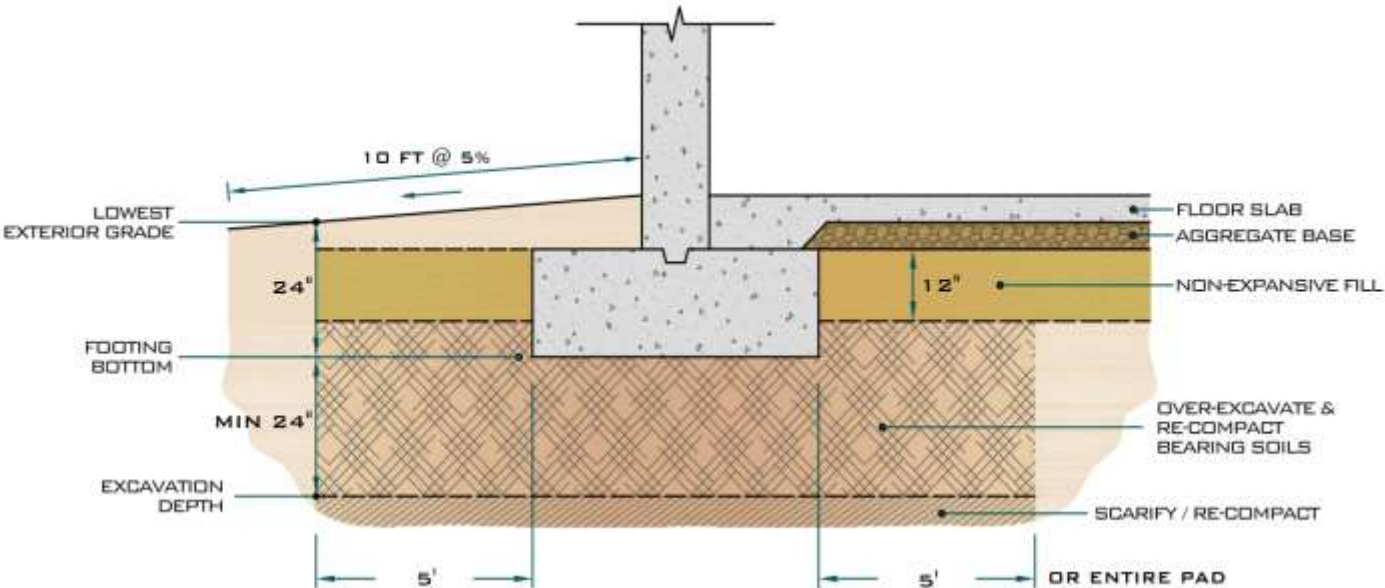


Figure 3.3.1 Foundation Detail

These bearing capacities refer to the total of all loads, dead and live, and are net pressures. They may be increased one-third for wind, seismic or other loads of short duration. These values may be increased by one-third as the allowable toe pressure for retaining walls. All footing excavations should be level and cleaned of all loose or disturbed materials. Positive drainage away from the proposed buildings **must** always be maintained.

Continuous wall footings and isolated rectangular footings should be designed with minimum widths of 16 and 24 inches respectively, regardless of the resultant bearing pressure. Lightly loaded interior partitions (less than 800 plf) may be supported on reinforced thickened slab sections (minimum 12 inches of bearing width).

Estimated settlements under design loads are on the order of ½ to 1-inch, virtually all of which will occur during construction. Post-construction differential settlements will be on the order of one-half the total settlement, under existing and compacted moisture contents. Additional localized settlements of the same magnitude could occur if native supporting soils were to experience a significant increase in moisture content. **Positive drainage away from structures and controlled routing of roof runoff must be provided and maintained to prevent ponding adjacent to perimeter walls.** Planters requiring heavy water should **not** be placed adjacent to or within 5 feet of the building. Care should be taken in design and construction to ensure that domestic and interior storm drain water is contained to prevent seepage. Roof drainage should be directed to paved areas or storm drains. They should not discharge into planters adjacent to the structures.

Continuous footings and stem walls should be reinforced to distribute stresses arising from small differential movements, and long walls should be provided with control joints to accommodate these movements. Reinforcement and frequent control joints are suggested to allow slight movement and prevent minor floor slab cracking, especially in floor areas covered with hard tile. **Due to the potential for post construction wall settlement if/when impacted by water, it is recommended to NOT tie the floors to the concrete tilt panel walls.** The floors should be allowed to “float” independent of the walls. When the closure pours, and first floor panel are reinforced and tied to the wall, normal concrete shrinkage causes the first free joint to open more than the others. Any settlement of the walls, even within the estimated tolerances, will magnify this opening. If it becomes necessary to tie the walls to the floor, a detail (such as dowels) to provide load transfer across this open joint is recommended.

3.4 Lateral Pressures

The following equivalent fluid lateral pressure values may be utilized for the proposed construction:

Active Pressures	
Unrestrained Walls	35 pcf
At-Rest Pressures	
Restrained Walls	55 pcf
Passive Pressures	
Continuous Footings	300 pcf
Spread Footings	350 pcf
Coefficient of Friction (w/ passive pressure)	0.35
Coefficient of Friction (w/out passive pressure)	0.45

All backfill must be compacted to not less than 95 percent (ASTM D-698) to mobilize these passive values at low strain. Expansive soils should not be used as retaining wall backfill, except as a surface seal to limit infiltration of storm/irrigation water. The expansive pressures could greatly increase active pressures.

3.5 Fill and Backfill

In-situ soils are considered suitable for use in general grading fills and engineered fills below foundations but should **not** be used in the top 12 inches of building pad fill, contiguous sidewalks or as retaining wall backfill. The top 12 inches of pad fill should be completed with an approved low or non-expansive soil, either approved imported common borrow or select granular soil. If select granular soils are used, the 4 inches of under-slab aggregate base may be included as part of the top 12 inches. Otherwise, 12 inches of approved common borrow should be used in addition to the normal 4 inches of aggregate base. Native soils can also be treated with lime. Refer to section 3.9 for additional details.

If imported common fill for use in site grading is required, it should be examined by a Soils Engineer to ensure that it is of low swell potential and free of organic or otherwise deleterious material. In general, the fill should have 100 percent passing the 3-inch sieve and no more than 50 percent passing the #200 sieve. For the fine fraction (passing the 40 sieve), the liquid limit and plasticity index should not exceed 30 percent and 10 percent, respectively. It should exhibit less than 1.5 percent swell potential when compacted to 95 percent of maximum dry density (ASTM D-698) at a moisture content of 2 percent below optimum, confined under a 100 psf surcharge, and inundated.

Fill should be placed on subgrade, which has been properly prepared and approved by a Soils Engineer. The fill must be wetted and thoroughly mixed to achieve optimum moisture content, ± 2 percent (optimum to 3 percent above optimum for under slab and sidewalk fill). Fill should be placed in horizontal lifts of 8-inch thickness (or as dictated by compaction equipment) and compacted to the percent of maximum dry density per ASTM D-698 set forth as follows:

A. Building Areas	
a. Below footing level	95
b. Below slabs-on-grade (non-expansive soils)	95
c. Below slabs-on-grade (expansive soils)	90-95 (max)
(Not recommended for the top 12 inches of Pad)	
B. Pavement Subgrade or Fill	
a. Below Asphalt Pavement	95
b. Below Concrete Pavement (Clay soils)	90-95 (max)
C. Utility Trench Backfill	95
D. Aggregate Base Course	
a. Below floor slabs	95
b. Below asphalt paving	100
E. Landscape Areas	90
F. Sidewalk Subgrade (Clay soils)	90-95 (max)

3.6 Utilities Installation

Trench excavations for shallow utilities can be accomplished by conventional trenching equipment. Excavation into the more cemented soils may require heavy equipment. The fact that a boring was advanced to a certain depth does not mean that the soils may be excavated by normal means. The excavating contractor must make their own assessment as to excavatability. Trench walls should stand near vertical for the short periods of time required to install shallow utilities, although some sloughing may occur in looser and/or sandier soils requiring laying back of side slopes and/or temporary shoring. Adequate precautions must be taken to protect workmen in accordance with all current governmental regulations.

Backfill of narrow utility trenches above bedding and initial backfill zones may be carried out with native excavated material. This material should be moisture-conditioned, placed in 8-inch lifts and mechanically compacted. Water settling is not recommended. Compaction requirements are summarized in the "Fill and Backfill" section of this report. Native soils do not meet the typical granular bedding and initial

backfill (see MAG Standard Specification Section 601.4.4) requirements of large diameter CMP tanks. These materials need to meet MAG Standard Specification Section 601 or the drainage engineers design and manufacture recommendations.

3.7 Slabs-On-Grade

To facilitate fine grading operations and aid in concrete curing, a 4-inch-thick layer of granular material conforming to the gradation for Aggregate Base (A.B.) as per M.A.G. Specification Section 702 should be utilized beneath the slab. Dried subgrade soils must be re-moistened prior to placing the aggregate base if allowed to dry out, especially if fine-grained soils are used in the top 12-inches of the pad.

For the support of slabs-on-grade, a Modulus of Subgrade Reaction (k), of 150 pci may be used for slabs supported on common non-expansive borrow. This may be increased to 250 pci (if required) for slabs supported on 12 inches of Well Graded Granular fill (MAG Spec 702 AB or Select Type B, ASTM D2940 Subbase for Roadways (or approved equal) or 8 inches cement/lime stabilized soil (+ 4 inches of aggregate base or 2 inches of washed ¾-inch rock). Common Borrow is NOT suitable for this higher modulus value. **See comments in Section 3.3 about not attaching the slab to the walls. It is highly recommended to consider smooth dowels, tapered dowels, or other means for load transfer across joints in floor slabs subject to heavy forklift or truck loads to eliminate the potential for joint failures.**

The native soil can store a significant amount of moisture, which could increase the natural vapor drive through the slab. Accordingly, if moisture sensitive flooring and/or adhesive are planned, the use of a vapor barrier directly under the slab is recommended. Vapor barriers should be a minimum 15-mil thick polyolefin (or equivalent), which meets ASTM E 1745 Class A specifications. Vapor barriers do increase the potential for slab curling and water entrapment under the slab. Accordingly, if a vapor barrier is used, additional precautions such as low slump concrete, frequent jointing and proper curing will be required to reduce curling potential and detailed to prevent the entrapment of outside water sources.

3.8 Asphalt/Concrete Pavement

If earthwork in paved areas is carried out to finish subgrade elevation as set forth herein, the subgrade will provide adequate support for pavements. The location designation is for reference only. The designer/owner should choose the appropriate sections to meet the anticipated traffic volume and life expectancy. The section capacity is reported as daily ESALs, Equivalent 18 kip Single Axle Loads. Typical heavy trucks impart 1.0 to 2.5 ESALs per truck depending on load. It takes approximately 1,200 passenger cars to impart 1 ESAL.

Table 3.8.1 Pavement Sections

Area of Placement	Flexible (AC Pavement)			Rigid (PCC Pavement)	
	Thickness		Daily 18-kip ESALs	Thickness	Daily 18-kip ESALs
	AC (0.39/0.42)	ABC (0.12)		PCCP	
Auto Parking	2.0"	6.0"	4	5.0"	14
Main Drives, Truck Parking, & Fire Lanes	3.0"	8.0"	32	6.0"	35
	4.0"	6.0"	70	7.0"	80
	4.5"	6.0"	121	8.0"	170
Notes: 1. Designs are based on AASHTO design equations and ADOT correlated R-Values. 2. The PCCP thickness is increased to provide better load transfer and reduce potential for joint & edge failures. Design PCCP per ACI 330R. 3. Full depth asphalt or increased asphalt thickness can be increased by adding 1.0-inch asphalt for each 3 inches of base course replaced.					

Pavement Design Parameters:

Assume:	One 18-kip Equivalent Single Axle Load (ESAL)/Truck
Life:	20 years
Subgrade Soil Profile:	
% Passing #200 sieve:	63 percent
Plasticity Index:	15 percent
k:	150 pci (assumed)
R-value:	25 (per ADOT tables)
M _R :	14,800 (per AASHTO design)

These designs assume that all subgrades are prepared in accordance with the recommendations contained in the "Site Preparation" and "Fill and Backfill" sections of this report, and paving operations carried out in a proper manner. If pavement subgrade preparation is not carried out immediately prior to paving, the entire area should be proof-rolled at that time with a heavy pneumatic-tired roller to identify locally unstable areas for repair.

Pavement base course material should be aggregate base per M.A.G. Section 702 Specifications. Asphalt concrete materials and mix design should conform to M.A.G. 710. It is recommended that a ½ inch or ¾ inch mix designation be used for the pavements. While a ¾ inch mix may have a somewhat

rougher texture, it offers more stability and resistance to scuffing, particularly in truck turning areas. Pavement installation should be carried out under applicable portions of M.A.G. Section 321 and municipality standards. The asphalt supplier should be informed of the pavement use and be required to provide a mix that will provide stability and be aesthetically acceptable. Some of the newer M.A.G. mixes are very coarse and could cause placing and finish problems. A mix design should be submitted for review to determine if it will be acceptable for the intended use.

For sidewalks and other areas not subjected to vehicular traffic a 4-inch section of concrete will be sufficient. For pavement areas with anticipated vehicular traffic a minimum section of 5 inches of concrete will provide adequate service. For trash and dumpster enclosures a thicker section of 6 inches of concrete is recommended.

Portland Cement Concrete Pavement must have a minimum 28-day flexural strength of 600 psi (compressive strength of approximately 4,000 psi). It may be cast directly on the prepared subgrade with proper compaction (reduced) and the elevated moisture content as recommended in the report. Lacking an aggregate base course, attention must be paid to using low slump concrete and proper curing, especially on the thinner sections. No reinforcement is necessary. Joint design and spacing should be in accordance with ACI recommendations. Construction joints should contain dowels or be tongue-and-grooved to provide load transfer. Tie bars are recommended on the joints adjacent to unsupported edges. Maximum joint spacing in feet should not exceed 2 to 3 times the thickness in inches. Joint sealing with a quality silicone sealer is recommended to prevent water from entering the subgrade, allowing pumping and loss of support. If joint sealing is not used, it is recommended to use a 4-inch aggregate subbase to reduce potential for loss of support due to water infiltration.

Proper subgrade preparation and joint sealing will reduce (but not eliminate) the potential for slab movements (thus cracking) on the expansive native soils. Frequent jointing will reduce uncontrolled cracking and increase the efficiency of aggregate interlock joint transfer.

It must be noted that all new asphalt pavements will eventually crack. Cracking in asphalt pavement is typical and should be expected over the life of the pavement. In fact, it has been our experience of late that the new asphalt binders that are available, we are seeing the onset or earlier aging and block shrinkage cracking. These require routine maintenance to prevent accelerated deterioration. Accordingly, it is highly recommended to establish a maintenance program where the cracks are routinely filled as they appear beginning at about the second year of life. It is also recommended that surface fog seal coats be considered beginning at about year 5 and every 5 years after. This will help preserve the pavements, extending the service life.

3.9 Lime Stabilization

As an alternative to placing 12 inches of non-expansive common borrow import to complete the building pad (and/or importing common borrow for engineered fill under the foundations), the native soils may be treated with chemical lime slurry to reduce the plasticity and swell potential to less than 1.5 percent. **A minimum of 8 inches of lime-stabilized soil (LSS) is recommended under slabs-on-grade plus 4 inches of ABC.** Commercial lime slurry is recommended to reduce the environmental concerns for blowing lime dust and to ensure proper hydration and mixing. Depending on the percentage of lime used and its reaction with the native clays, cementing action could increase the subgrade support capacity, improving the load carrying capacity of the floor slab. The specified under-slab aggregate base course should not be eliminated.

Lime may also be used to stabilize subgrade soils in vehicle drive and parking areas provided that a mix design is performed to determine if the native soils will react with the lime. There can be a substantial reduction in aggregate base thickness (up to full replacement) for a given traffic volume when the subgrade is treated with lime. Correspondingly, the capacity can be greatly increased. There have been reports of pavement surfacing cracking on thin pavement sections where lime is used. Accordingly, proper subgrade preparation below the lime treated soils will be required to reduce (not eliminate) this potential. This includes increasing the moisture content to at least optimum to 3 percent above and limiting compaction to 90 to 95 percent to a depth of at least 6 inches. This can be accomplished in-place by deep mixing, or if not possible with the equipment used, removing the layer to be treated and preparing the subgrade prior to placing the stabilized base course. The surface of the treated grade should be sealed with an asphalt prime coat to act as a curing agent and prevent the surface from drying prior to installing the pavement. It is also recommended to provide at least 2+ inches on AB on top of the LSS prior to paving. This will provide a layer of unbound material that can be used to fine grade the parking lot (less re-working of the LSS) and to provide an interlayer to further reduce the reflective cracking. A minimum LSS section of 6 inches is recommended.

The application of lime and soil stabilization should be carried out by a contractor experienced in this type of work. Consideration may be given to the use of a slurry rather than dry application since dusting is a concern. Use of a lime slurry will also reduce the amount of loss and provide consistent coverage. This technology is locally available and is currently being used on recent projects. **A mix design study would be required to determine the amount of lime required** to reduce plasticity and/or increase strength. All lime stabilization work should be carried out in accordance with M.A.G. Section 309 **including the minimum 5 percent lime content to prevent reversion.**

4.0 GENERAL

The scope of this investigation and report includes only regional published considerations for seismic activity and ground fissures resulting from subsidence due to groundwater withdrawal, not any site-specific studies. The scope does not include any considerations of hazardous releases or toxic contamination of any type.

Our analysis of data and the recommendations presented herein assume that soil conditions do not vary significantly from those found at specific sample locations. Our work has been performed in accordance with generally accepted engineering principles and practice; this warranty is in lieu of all other warranties expressed or implied.

We recommend that a representative of the Geotechnical Engineer observe and test the earthwork and foundation portions of this project to ensure compliance to project specifications and the field applicability of subsurface conditions which are the basis of the recommendations presented in this report. If any significant changes are made in the scope of work or type of construction that was assumed in this report, we must review such revised conditions to confirm our findings if the conclusions and recommendations presented herein are to apply.

Respectfully submitted,
SPEEDIE & ASSOCIATES, LLC



Juan C. Salas



Keith R. Gravel, P.E.



APPENDIX

FIELD AND LABORATORY INVESTIGATION

SOIL BORING LOCATION PLAN

SOIL LEGEND

LOG OF TEST BORINGS

TABULATION OF TEST DATA

CONSOLIDATION TEST

MOISTURE-DENSITY RELATIONS

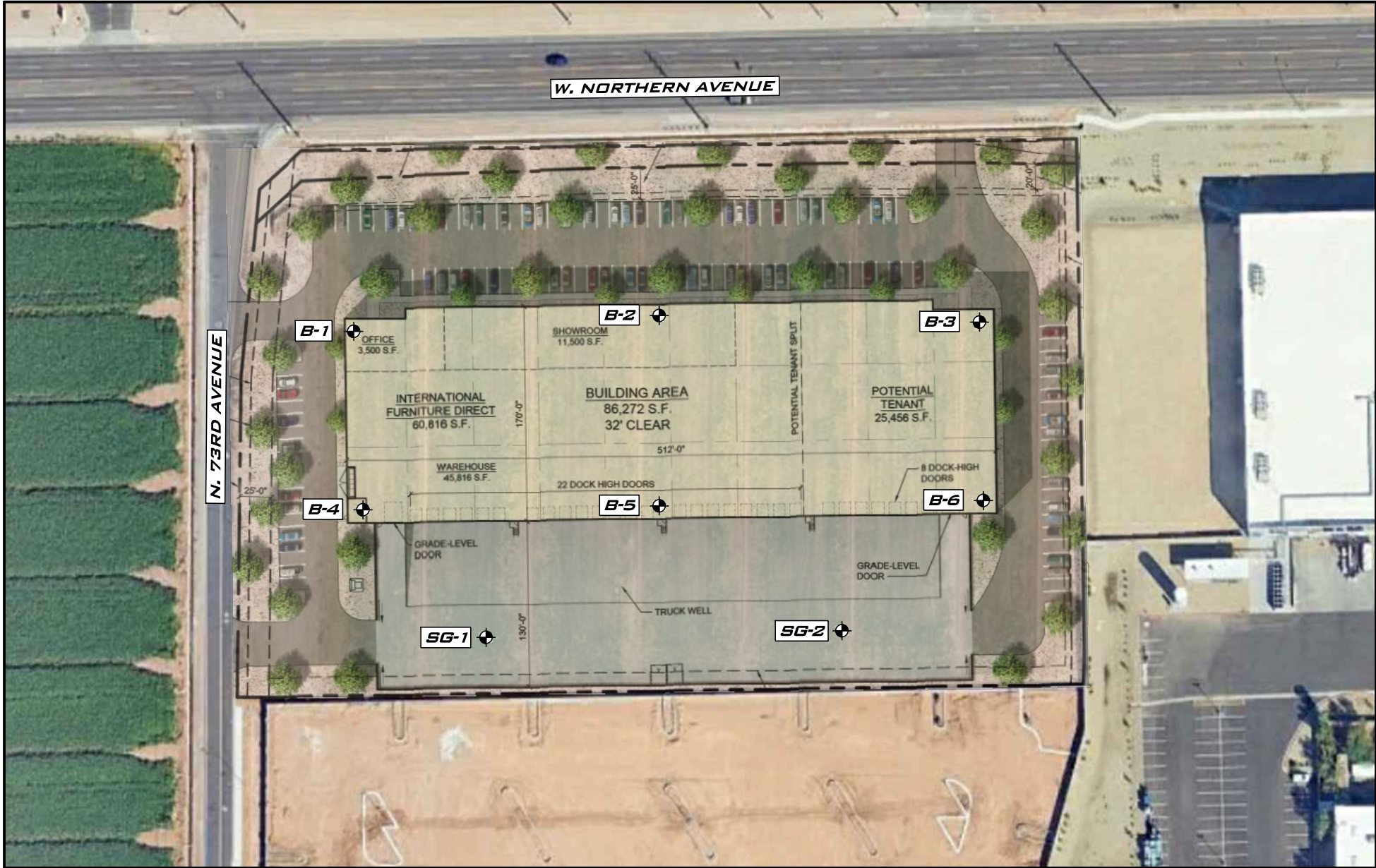
SWELL TEST DATA



FIELD AND LABORATORY INVESTIGATION

On October 22, 2024, soil test borings were drilled at the approximate locations shown on the attached Soil Boring Location Plan. All exploration work was carried out under the full-time supervision of our staff geologist, who recorded subsurface conditions and obtained samples for laboratory testing. The soil borings were advanced with a truck-mounted CME-75 drill rig utilizing 7-inch diameter hollow stem flight augers. Detailed information regarding the borings and samples obtained can be found on an individual Log of Test Boring prepared for each drilling location.

Laboratory testing consisted of moisture content, dry density, grain-size distribution, and plasticity (Atterberg Limits) tests for classification and pavement design parameters. Remolded swell tests were performed on samples compacted to densities and moisture contents expected during construction. Compression tests were performed on a selected ring sample to estimate settlements and determine effects of inundation. All field and laboratory data are presented in this appendix.



⬮ - APPROXIMATE SOIL BORING LOCATIONS



SOIL BORING LOCATION PLAN

INTERNATIONAL FURNITURE DIRECT
SEC 73RD AVENUE & NORTHERN AVENUE
GLENDALE, ARIZONA

**SPEEDIE
AND ASSOCIATES**
A UES Company

SOIL LEGEND

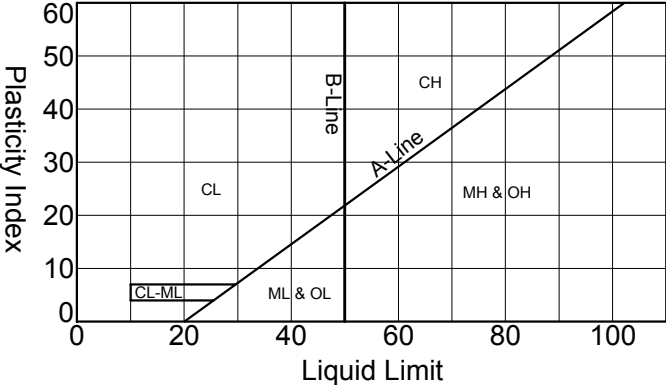
SAMPLE DESIGNATION		DESCRIPTION	
	AS	Auger Sample	A grab sample taken directly from auger flights.
	BS	Large Bulk Sample	A grab sample taken from auger spoils or from bucket of backhoe.
	S	Spoon Sample	Standard Penetration Test (ASTM D-1586) Driving a 2.0 inch outside diameter split spoon sampler into undisturbed soil for three successive 6-inch increments by means of a 140 lb. weight free falling through a distance of 30 inches. The cumulative number of blows for the final 12 inches of penetration is the Standard Penetration Resistance.
	RS	Ring Sample	Driving a 3.0 inch outside diameter spoon equipped with a series of 2.42-inch inside diameter, 1-inch long brass rings, into undisturbed soil for one 12-inch increment by the same means of the Spoon Sample. The blows required for the 12 inches of penetration are recorded.
	LS	Liner Sample	Standard Penetration Test driving a 2.0-inch outside diameter split spoon equipped with two 3-inch long, 3/8-inch inside diameter brass liners, separated by a 1-inch long spacer, into undisturbed soil by the same means of the Spoon Sample.
	ST	Shelby Tube	A 3.0-inch outside diameter thin-walled tube continuously pushed into the undisturbed soil by a rapid motion, without impact or twisting (ASTM D-1587).
	--	Continuous Penetration Resistance	Driving a 2.0-inch outside diameter "Bullnose Penetrometer" continuously into undisturbed soil by the same means of the spoon sample. The blows for each successive 12-inch increment are recorded.

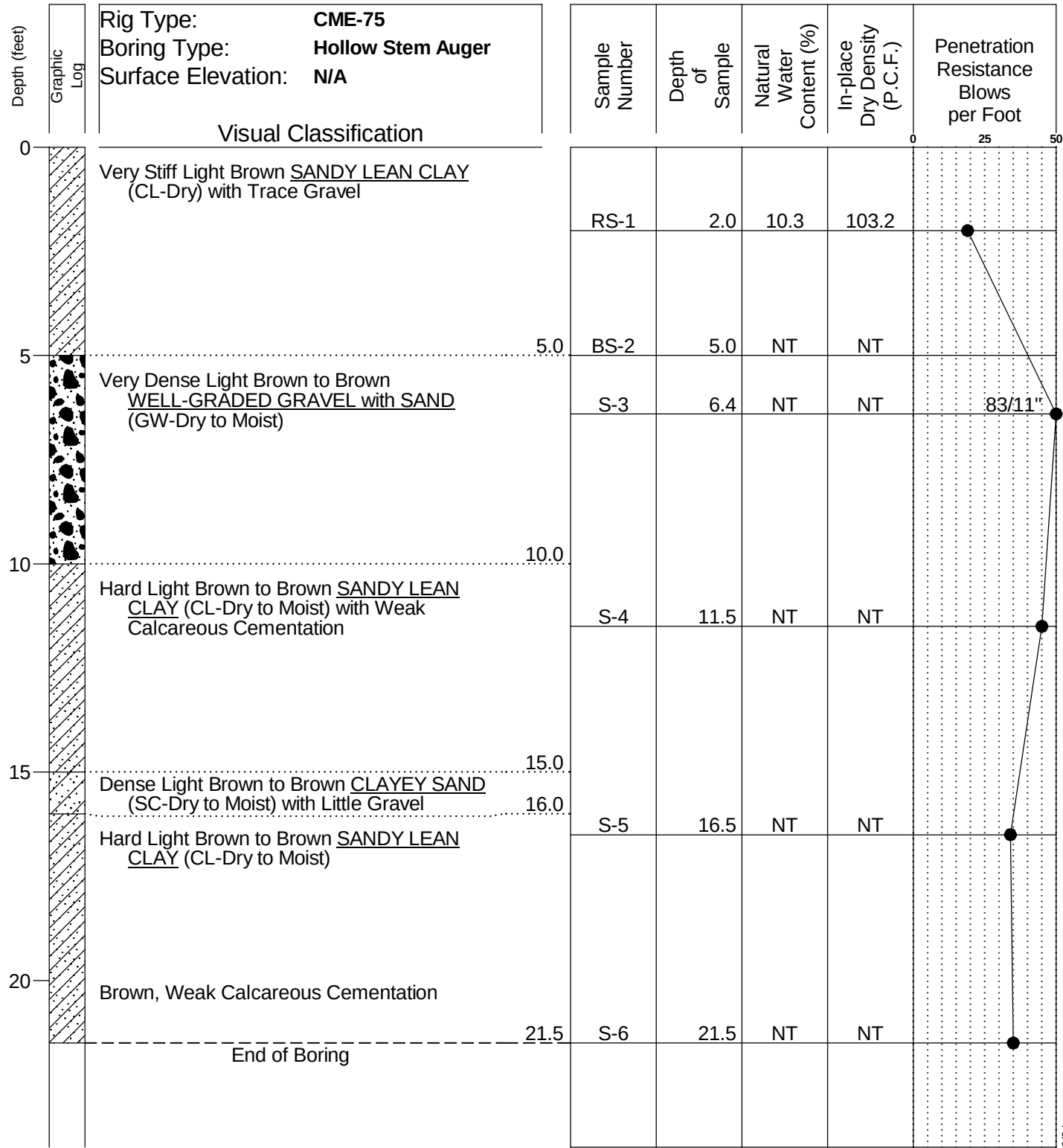
CONSISTENCY			RELATIVE DENSITY	
Clays & Silts	Blows/Foot	Strength (tons/sq ft)	Sands & Gravels	Blows/Foot
Very Soft	0 - 2	0 - 0.25	Very Loose	0 - 4
Soft	2 - 4	0.25 - 0.5	Loose	5 - 10
Firm	5 - 8	0.5 - 1.0	Medium Dense	11 - 30
Stiff	9 - 15	1 - 2	Dense	31 - 50
Very Stiff	16 - 30	2 - 4	Very Dense	> 50
Hard	> 30	> 4		

MAJOR DIVISIONS			SYMBOLS		TYPICAL DESCRIPTIONS
			GRAPH	LETTER	
COARSE GRAINED SOILS	GRAVEL AND GRAVELLY SOILS	CLEAN GRAVELS (LITTLE OR NO FINES)		GW	WELL-GRADED GRAVELS, GRAVEL - SAND MIXTURES, LITTLE OR NO FINES
		GRAVELS WITH FINES (APPRECIABLE AMOUNT OF FINES)		GP	POORLY-GRADED GRAVELS, GRAVEL - SAND MIXTURES, LITTLE OR NO FINES
		GRAVELS WITH FINES (APPRECIABLE AMOUNT OF FINES)		GM	SILTY GRAVELS, GRAVEL - SAND - SILT MIXTURES
	SAND AND SANDY SOILS	CLEAN SANDS (LITTLE OR NO FINES)		GC	CLAYEY GRAVELS, GRAVEL - SAND - CLAY MIXTURES
		CLEAN SANDS (LITTLE OR NO FINES)		SW	WELL-GRADED SANDS, GRAVELLY SANDS, LITTLE OR NO FINES
		SANDS WITH FINES (APPRECIABLE AMOUNT OF FINES)		SP	POORLY-GRADED SANDS, GRAVELLY SAND, LITTLE OR NO FINES
FINE GRAINED SOILS	SILTS AND CLAYS	CLEAN SANDS (LITTLE OR NO FINES)		SM	SAND - SILT MIXTURES
	SANDS WITH FINES (APPRECIABLE AMOUNT OF FINES)	SANDS WITH FINES (APPRECIABLE AMOUNT OF FINES)		SC	CLAYEY SANDS, SAND - CLAY MIXTURES
	SANDS WITH FINES (APPRECIABLE AMOUNT OF FINES)	SANDS WITH FINES (APPRECIABLE AMOUNT OF FINES)			
	SILTS AND CLAYS	Liquid Limit LESS THAN 50		ML	INORGANIC SILTS AND VERY FINE SANDS, ROCK FLOUR, SILTY OR CLAYEY FINE SANDS OR CLAYEY SILTS WITH SLIGHT PLASTICITY
	SILTS AND CLAYS	Liquid Limit LESS THAN 50		CL	INORGANIC CLAYS OF LOW TO MEDIUM PLASTICITY, GRAVELLY CLAYS, SANDY CLAYS, SILTY CLAYS, LEAN CLAYS
	SILTS AND CLAYS	Liquid Limit LESS THAN 50		OL	ORGANIC SILTS AND ORGANIC SILTY CLAYS OF LOW PLASTICITY
HIGHLY ORGANIC SOILS	SILTS AND CLAYS	Liquid Limit GREATER THAN 50		MH	INORGANIC SILTS, MICACEOUS OR DIATOMACEOUS FINE SAND OR SILTY SOILS
	SILTS AND CLAYS	Liquid Limit GREATER THAN 50		CH	INORGANIC CLAYS OF HIGH PLASTICITY
	SILTS AND CLAYS	Liquid Limit GREATER THAN 50		OH	ORGANIC CLAYS OF MEDIUM TO HIGH PLASTICITY, ORGANIC SILTS
	SILTS AND CLAYS	Liquid Limit GREATER THAN 50			
	SILTS AND CLAYS	Liquid Limit GREATER THAN 50			
	SILTS AND CLAYS	Liquid Limit GREATER THAN 50		PT	PEAT, HUMUS, SWAMP SOILS WITH HIGH ORGANIC CONTENTS

NOTE: DUAL OR MODIFIED SYMBOLS MAY BE USED TO INDICATE BORDERLINE SOIL CLASSIFICATIONS OR TO PROVIDE A BETTER GRAPHICAL PRESENTATION OF THE SOIL

MATERIAL SIZE	PARTICLE SIZE			
	Lower Limit		Upper Limit	
	mm	Sieve Size ♦	mm	Sieve Size ♦
SANDS				
Fine	0.075	#200	0.42	#40
Medium	0.420	#40	2.00	#10
Coarse	2.000	#10	4.75	#4
GRAVELS				
Fine	4.75	#4	19	0.75" x
Coarse	19	0.75" x	75	3" x
COBBLES	75	3" x	300	12" x
BOULDERS	300	12" x	900	36" x
♦U.S. Standard		xClear Square Openings		





Boring Date: 10-22-24

Field Engineer/Technician: E. Williams

Driller: F. Dominguez

Contractor: Resilient Drilling

Water Level		
Depth	Hour	Date
Free Water was Not Encountered		

NT = Not Tested

SPEEDIE AND ASSOCIATES

Log of Test Boring Number: **B- 1**

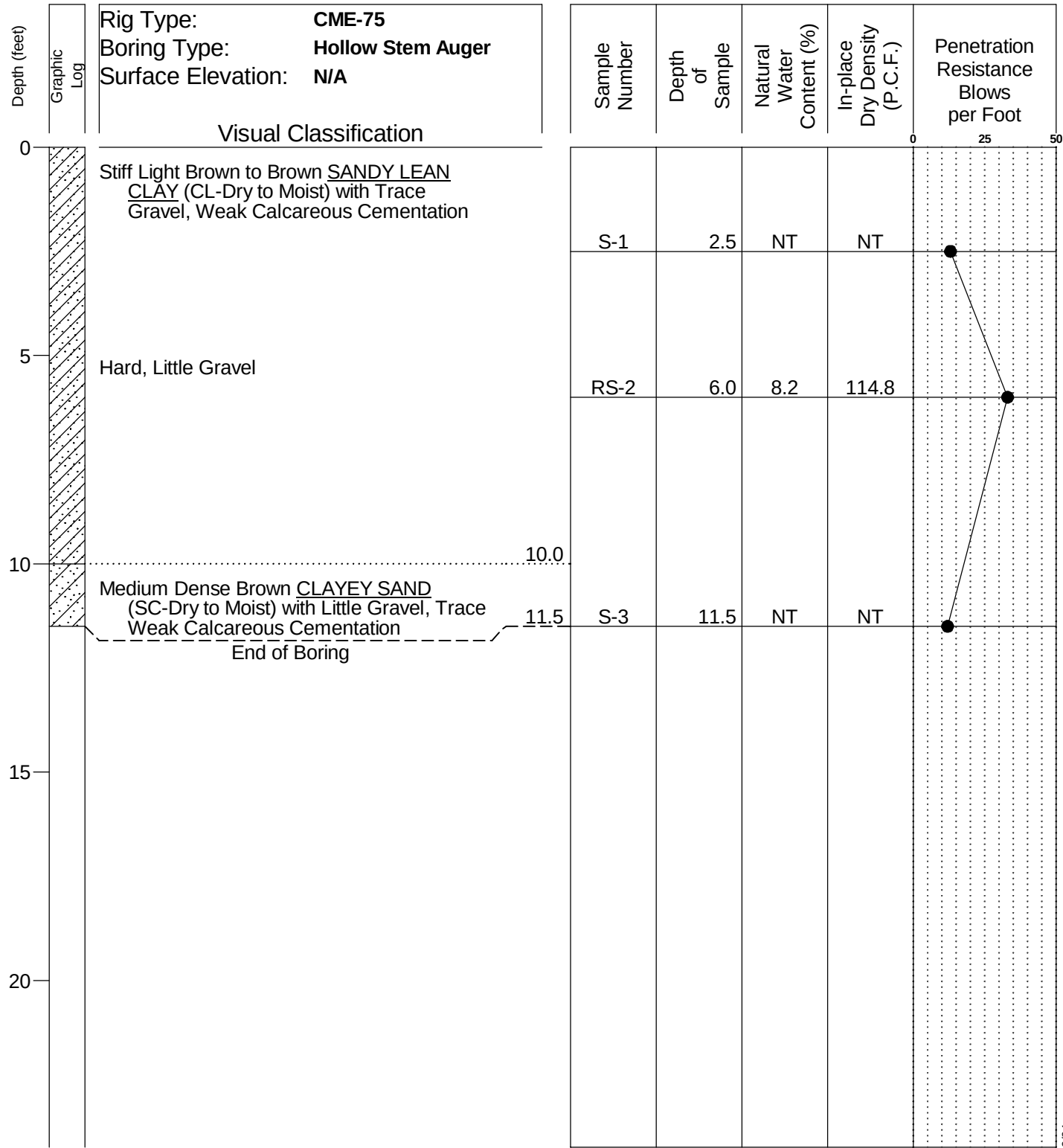
International Furniture Direct

SEC 73rd Avenue & Northern Avenue

Glendale, Arizona

Project No.: 242091SA

SPEEDIE 242091SA.GPJ GENGEO.GDT 11/12/24



Boring Date: 10-22-24

Field Engineer/Technician: E. Williams

Driller: F. Dominguez

Contractor: Resilient Drilling

Water Level		
Depth	Hour	Date
Free Water was Not Encountered		

NT = Not Tested

SPEEDIE AND ASSOCIATES

Log of Test Boring Number: **B- 2**

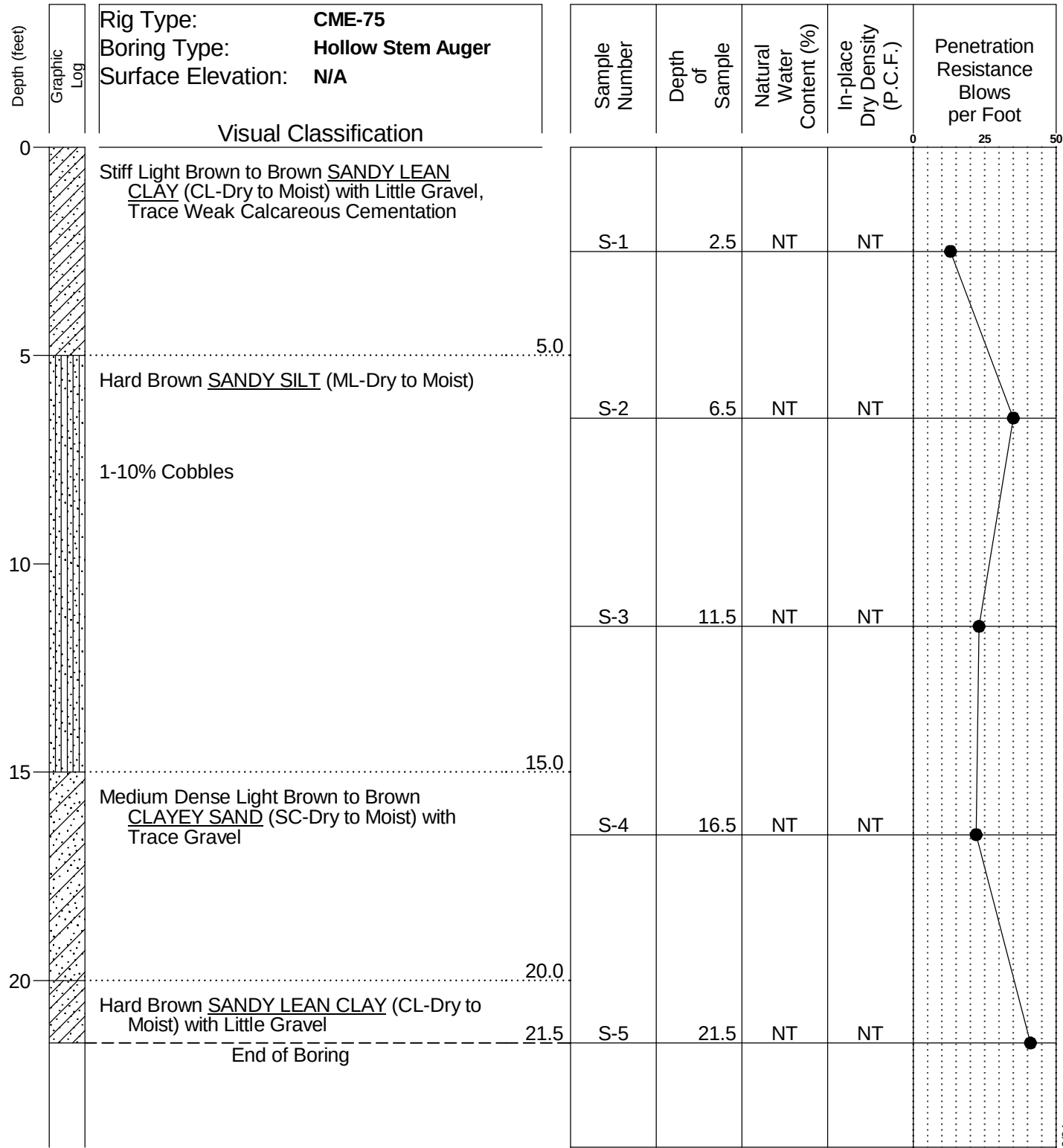
International Furniture Direct

SEC 73rd Avenue & Northern Avenue

Glendale, Arizona

Project No.: 242091SA

SPEEDIE 242091SA.GPJ GENGEO.GDT 11/12/24



Boring Date: 10-22-24
Field Engineer/Technician: E. Williams
Driller: F. Dominguez
Contractor: Resilient Drilling

Water Level		
Depth	Hour	Date
Free Water was Not Encountered		

NT = Not Tested

SPEEDIE AND ASSOCIATES

Log of Test Boring Number: **B- 3**

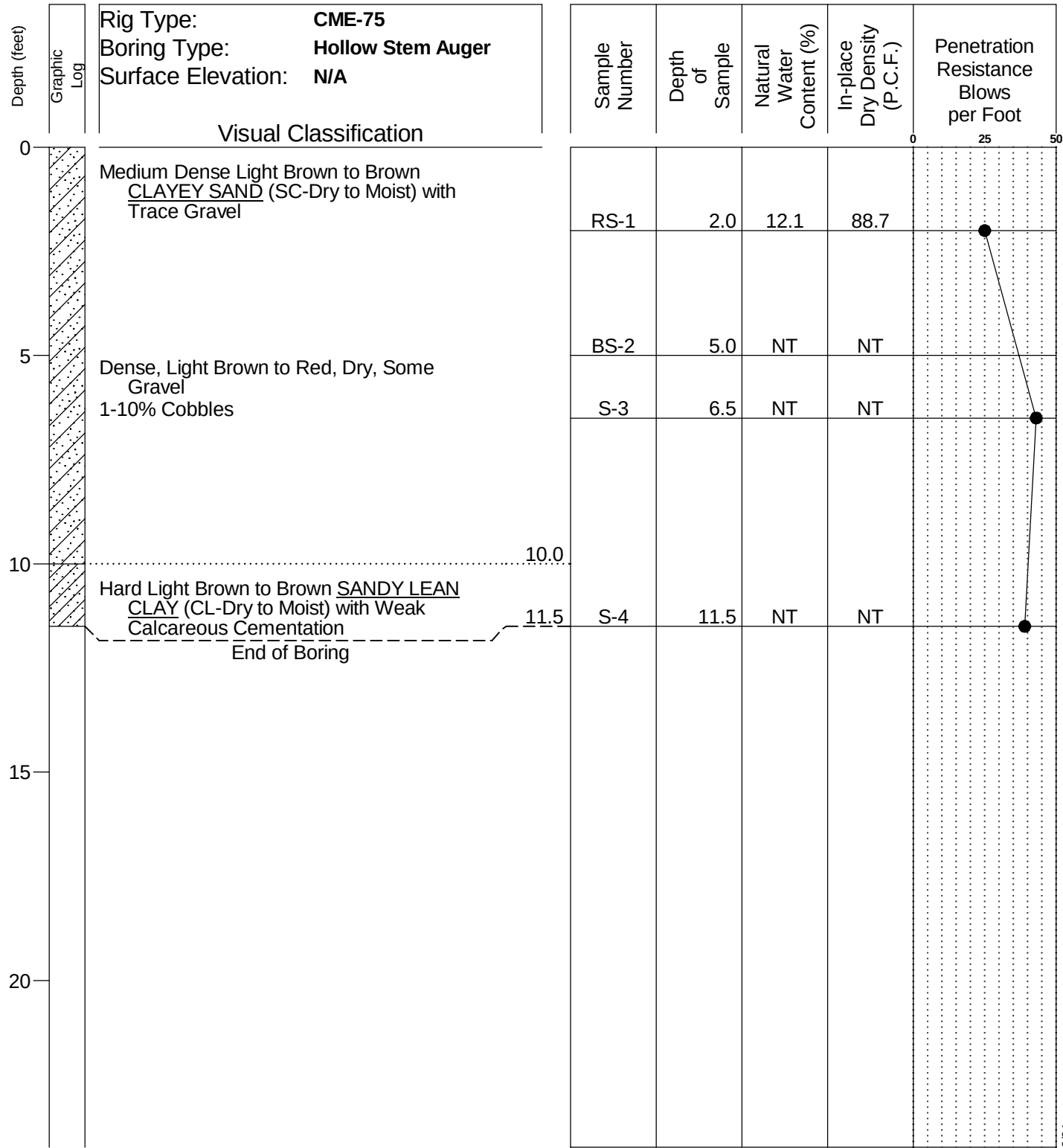
International Furniture Direct

SEC 73rd Avenue & Northern Avenue

Glendale, Arizona

Project No.: 242091SA

SPEEDIE 242091SA.GPJ GENGEO.GDT 11/12/24



Boring Date: 10-22-24

Field Engineer/Technician: E. Williams

Driller: F. Dominguez

Contractor: Resilient Drilling

Water Level		
Depth	Hour	Date
Free Water was Not Encountered		

NT = Not Tested

SPEEDIE
AND ASSOCIATES

Log of Test Boring Number: **B- 4**

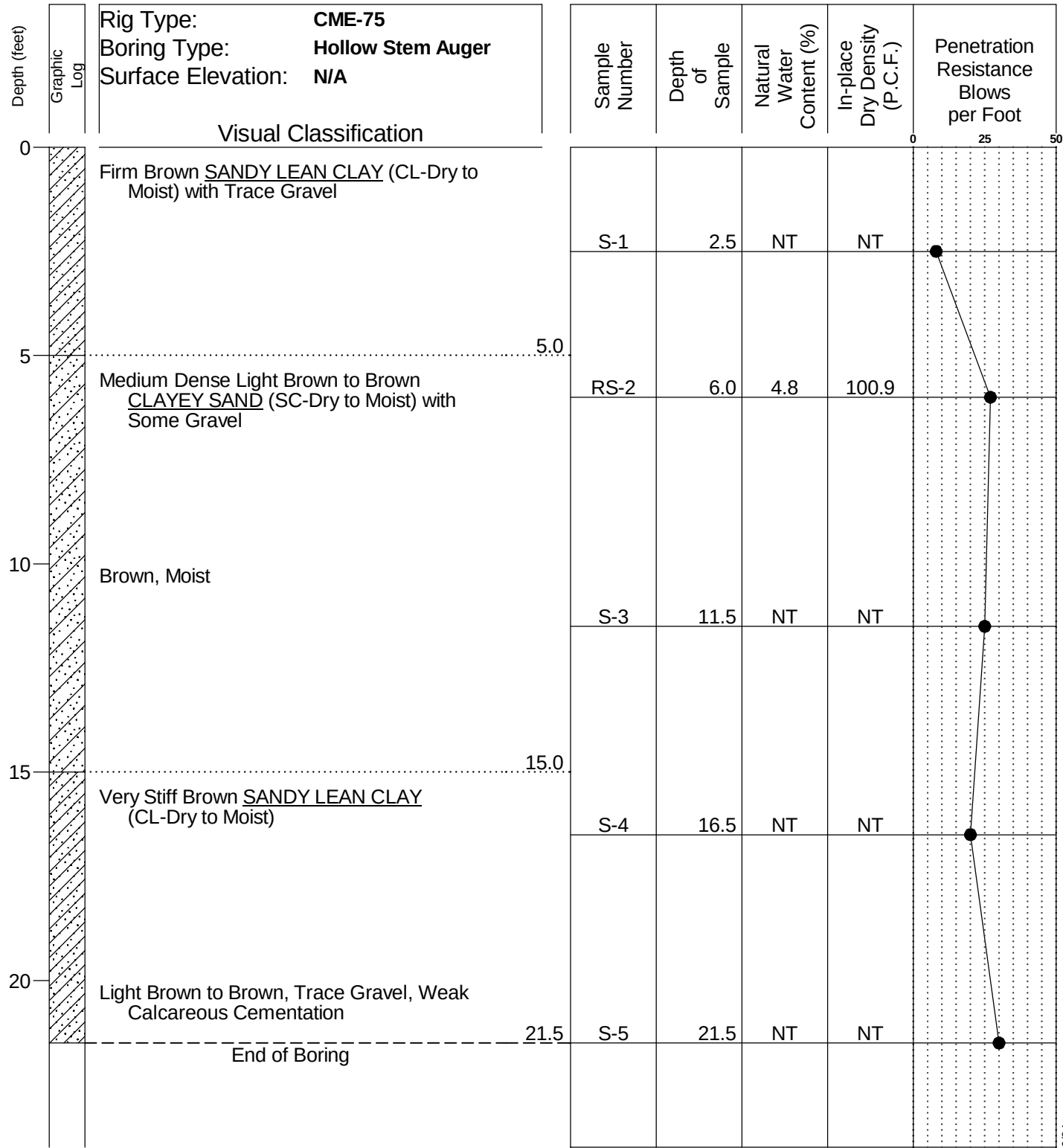
International Furniture Direct

SEC 73rd Avenue & Northern Avenue

Glendale, Arizona

Project No.: **242091SA**

SPEEDIE 242091SA.GPJ GENGEO.GDT 11/12/24



Boring Date: 10-22-24
Field Engineer/Technician: E. Williams
Driller: F. Dominguez
Contractor: Resilient Drilling

Water Level		
Depth	Hour	Date
Free Water was Not Encountered		

NT = Not Tested

SPEEDIE AND ASSOCIATES

Log of Test Boring Number: **B- 5**

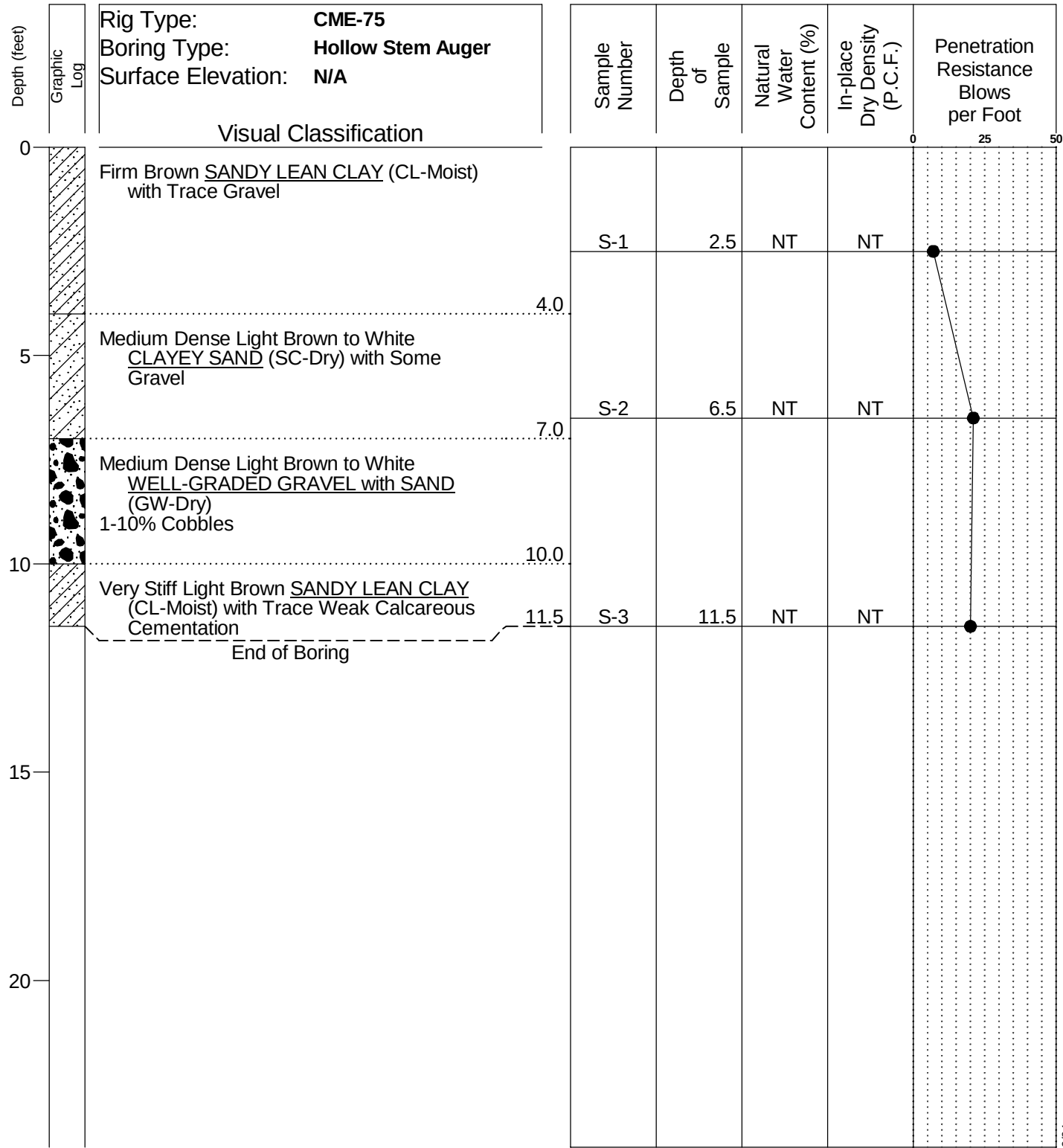
International Furniture Direct

SEC 73rd Avenue & Northern Avenue

Glendale, Arizona

Project No.: 242091SA

SPEEDIE 242091SA.GPJ GENGEO.GDT 11/20/24



Boring Date: 10-22-24

Field Engineer/Technician: E. Williams

Driller: F. Dominguez

Contractor: Resilient Drilling

Water Level		
Depth	Hour	Date
Free Water was Not Encountered		

NT = Not Tested

SPEEDIE AND ASSOCIATES

Log of Test Boring Number: **B- 6**

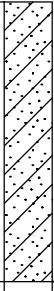
International Furniture Direct

SEC 73rd Avenue & Northern Avenue

Glendale, Arizona

Project No.: 242091SA

SPEEDIE 242091SA.GPJ GENGEO.GDT 11/12/24

Depth (feet)	Graphic Log	Rig Type: CME-75 Boring Type: Hollow Stem Auger Surface Elevation: N/A	Sample Number	Depth of Sample	Natural Water Content (%)	In-place Dry Density (P.C.F.)	Penetration Resistance Blows per Foot
		Visual Classification					
0		Brown <u>CLAYEY SAND</u> (SC-Dry to Moist) with Trace Gravel					
5		End of Test Pit	SG-1	5.0	NT	NT	
10							
15							
20							

Boring Date: 10-22-24
Field Engineer/Technician: E. Williams
Driller: F. Dominguez
Contractor: Resilient Drilling

Water Level		
Depth	Hour	Date
Free Water was Not Encountered		

NT = Not Tested

SPEEDIE AND ASSOCIATES

Log of Test Boring Number: SG- 1

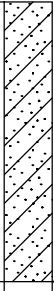
International Furniture Direct

SEC 73rd Avenue & Northern Avenue

Glendale, Arizona

Project No.: 242091SA

SPEEDIE 242091SA.GPJ GENGEO.GDT 11/12/24

Depth (feet)	Graphic Log	Rig Type: CME-75 Boring Type: Hollow Stem Auger Surface Elevation: N/A	Sample Number	Depth of Sample	Natural Water Content (%)	In-place Dry Density (P.C.F.)	Penetration Resistance Blows per Foot
		Visual Classification					
0		Brown <u>CLAYEY SAND</u> (SC-Dry to Moist) with Trace Gravel					
5		End of Test Pit	SG-1	5.0	NT	NT	
10							
15							
20							

Boring Date: 10-22-24
Field Engineer/Technician: E. Williams
Driller: F. Dominguez
Contractor: Resilient Drilling

Water Level		
Depth	Hour	Date
Free Water was Not Encountered		

NT = Not Tested

SPEEDIE AND ASSOCIATES

Log of Test Boring Number: SG- 2

International Furniture Direct

SEC 73rd Avenue & Northern Avenue

Glendale, Arizona

Project No.: 242091SA

SPEEDIE 242091SA.GPJ GEN GEO.GDT 11/12/24

TABULATION OF TEST DATA

SOIL BORING or TEST PIT NUMBER	SAMPLE NUMBER	SAMPLE TYPE	SAMPLE INTERVAL (ft)	NATURAL WATER CONTENT (Percent of Dry Weight)	IN-PLACE DRY DENSITY (Pounds Per Cubic Foot)	PARTICLE SIZE DISTRIBUTION (Percent Finer)					ATTERBERG LIMITS			UNIFIED SOIL CLASSIFICATION	SPECIMEN DESCRIPTION
						#200 SIEVE	#40 SIEVE	#10 SIEVE	#4 SIEVE	3" SIEVE	LIQUID LIMIT	PLASTIC LIMIT	PLASTICITY INDEX		
B- 1	RS-1	RING	1.0 - 2.0	10.3	103.2	50.1	72	88	93	100	26	16	10	CL	SANDY LEAN CLAY
B- 2	RS-2	RING	5.0 - 6.0	8.2	114.8	NT	NT	NT	NT	NT	NT	NT	NT		
B- 4	BS-2	BULK	0.0 - 5.0	NT	NT	49.6	69	89	95	100	38	20	18	SC	CLAYEY SAND
B- 4	RS-1	RING	1.0 - 2.0	12.1	88.7	55.7	74	90	96	100	35	19	16	CL	SANDY LEAN CLAY
B- 5	RS-2	RING	5.0 - 6.0	4.8	100.9	23.8	37	59	71	100	36	20	16	SC	CLAYEY SAND with GRAVEL

Sieve analysis results do not include material greater than 3". Refer to the actual boring logs for the possibility of cobble and boulder sized materials.

NT=Not Tested
Sheet 1 of 1

International Furniture Direct
SEC 73rd Avenue & Northern Avenue
Glendale, Arizona
Project No. 242091SA



CONSOLIDATION TEST

PROJECT: International Furniture Direct

PROJECT NO.: 242091SA

LOCATION: SEC 73rd Avenue & Northern Avenue

DATE: 10/22/24

BORING NO.: B- 1

SAMPLE NO.: RS-1

SAMPLE DEPTH: 1 to 2

LABORATORY NO.:

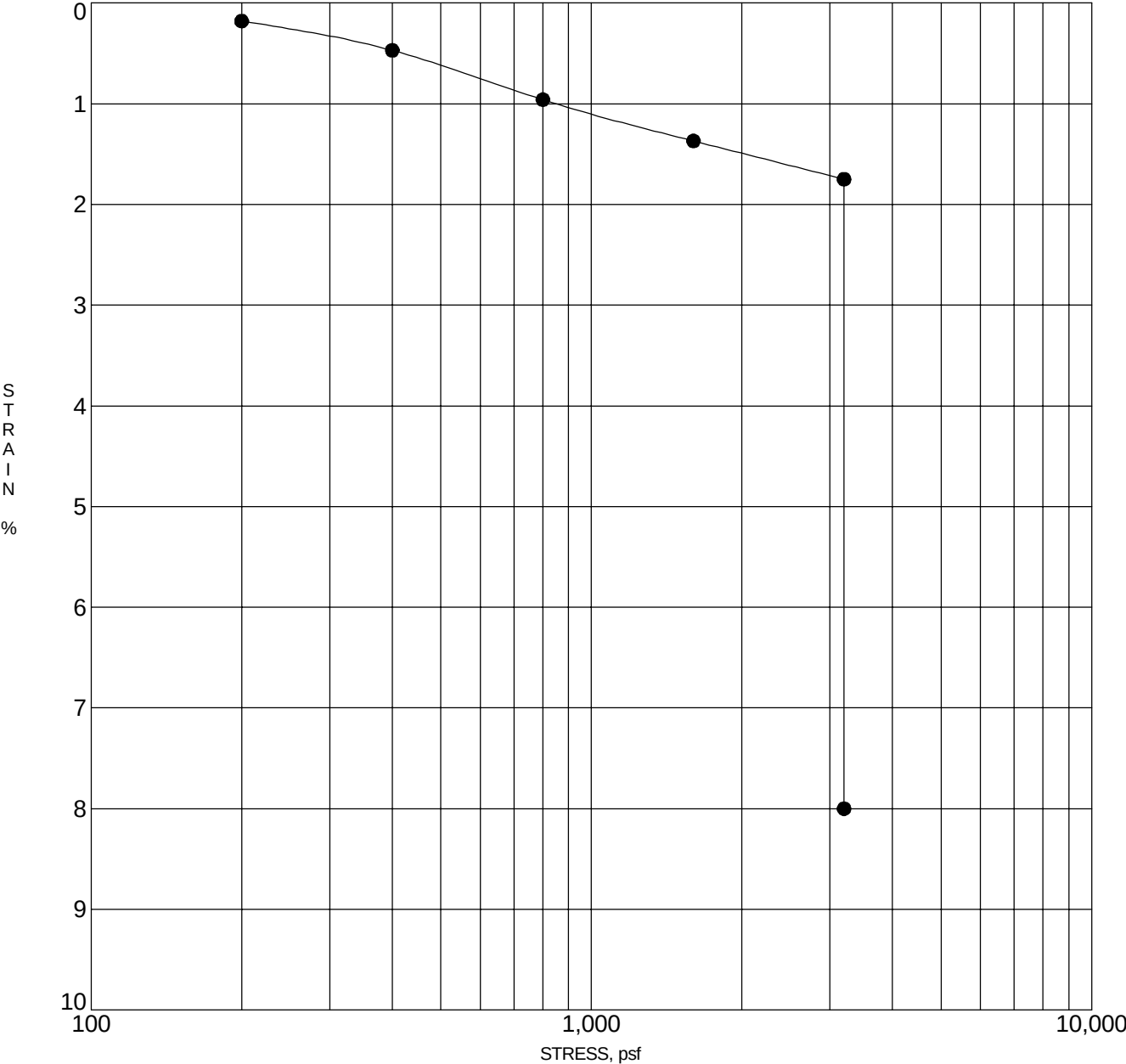
LIQUID LIMIT: 26

PLASTIC LIMIT: 16

PLASTICITY INDEX: 10

CLASSIFICATION: CL

ASTM SOIL DESCRIPTION: SANDY LEAN CLAY



Sample inundated at end of test at 3200 psf

CONSOLIDATION TEST

PROJECT: International Furniture Direct

PROJECT NO.: 242091SA

LOCATION: SEC 73rd Avenue & Northern Avenue

DATE: 10/22/24

BORING NO.: B- 4

SAMPLE NO.: RS-1

SAMPLE DEPTH: 1 to 2

LABORATORY NO.:

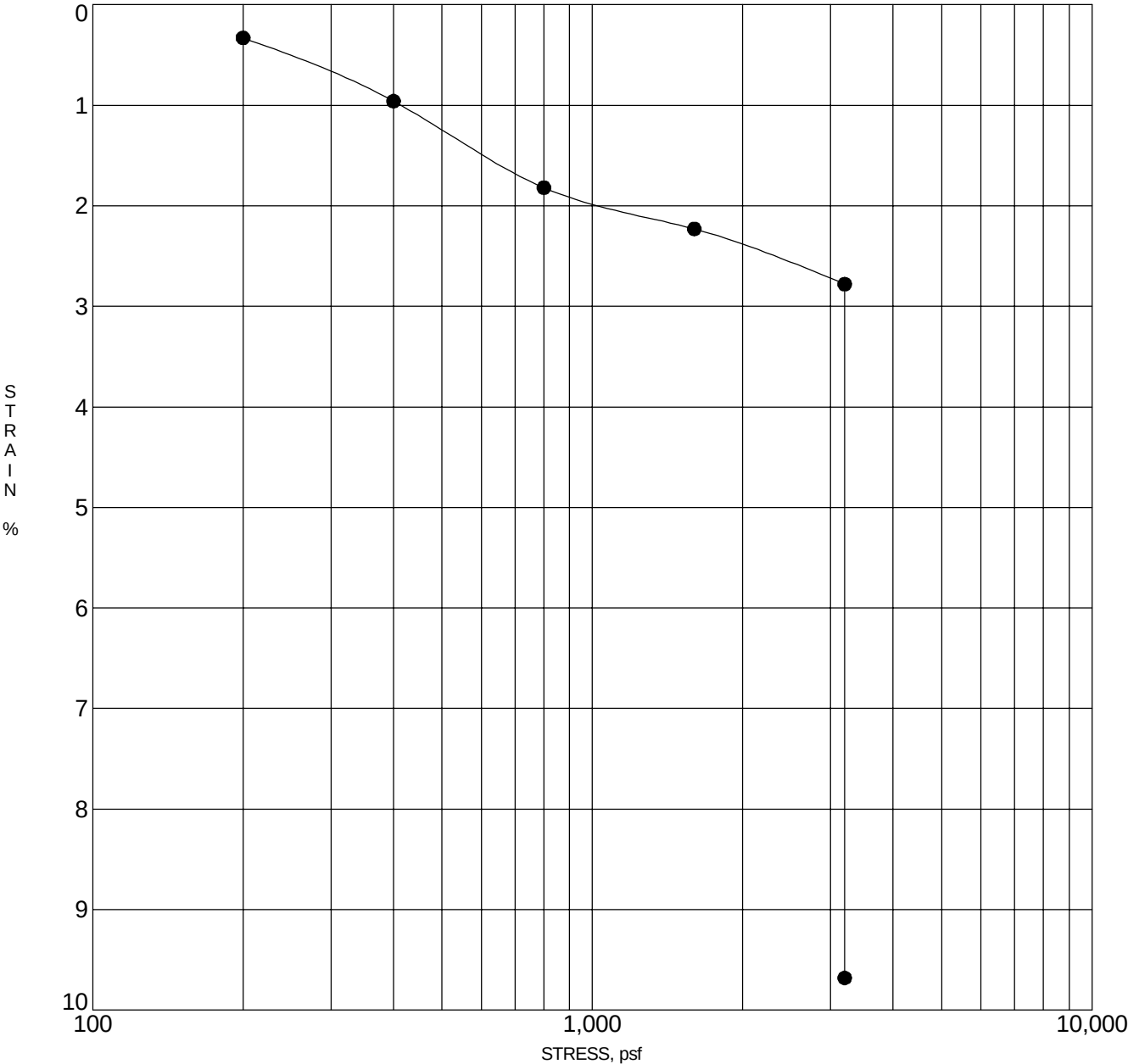
LIQUID LIMIT: 35

PLASTIC LIMIT: 19

PLASTICITY INDEX: 16

CLASSIFICATION: CL

ASTM SOIL DESCRIPTION: SANDY LEAN CLAY



MOISTURE-DENSITY RELATIONS

PROJECT: International Furniture Direct

PROJECT NO.: 242091SA

LOCATION: SEC 73rd Avenue & Northern Avenue

DATE: 10/22/24

BORING NO.: B- 4

SAMPLE NO.: BS-2

SAMPLE DEPTH: 0 to 5

LABORATORY NO.:

METHOD OF COMPACTION: D698A

LIQUID LIMIT: 38

PLASTIC LIMIT: 20

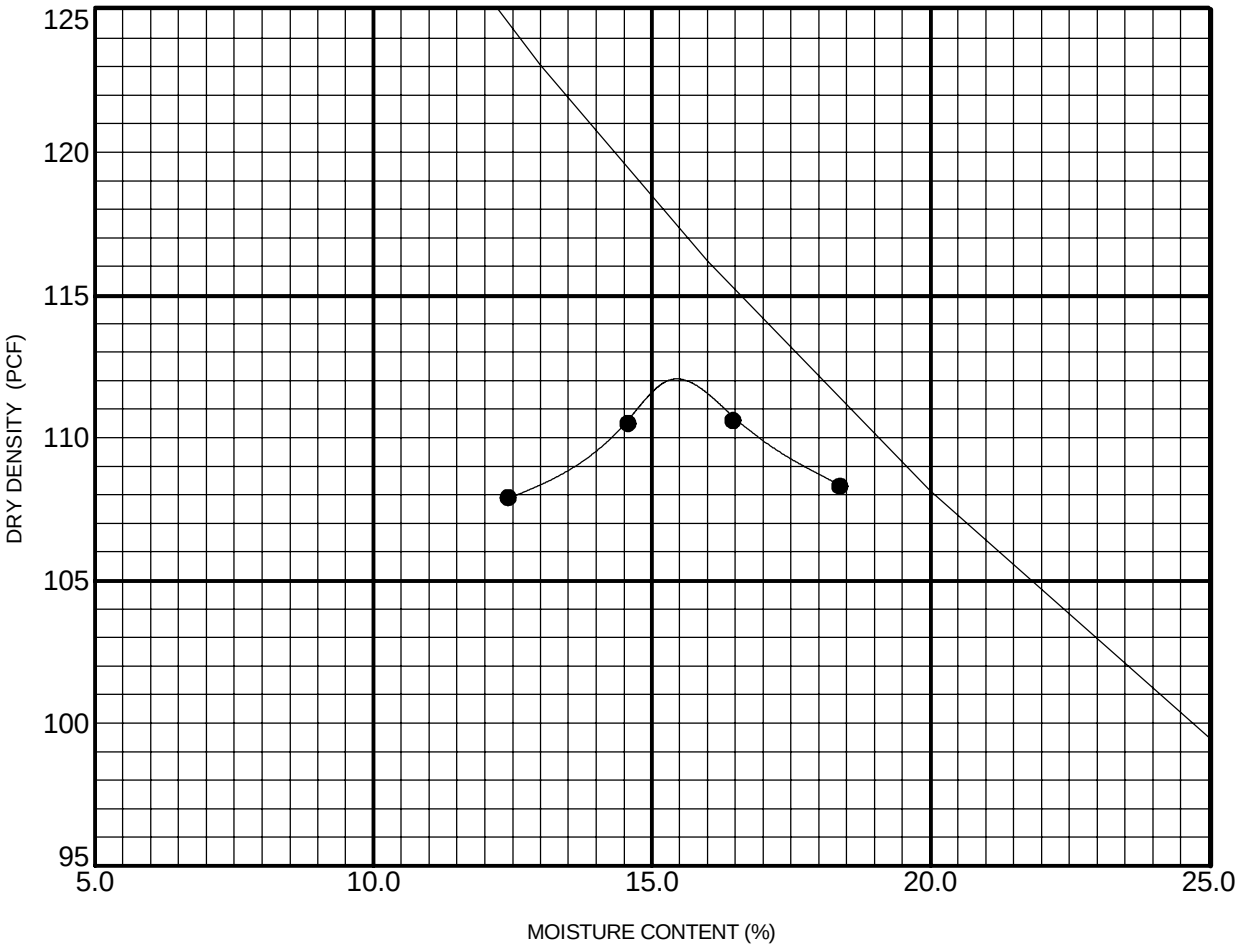
PLASTICITY INDEX: 17

CLASSIFICATION: SC

ASTM SOIL DESCRIPTION: CLAYEY SAND

MAXIMUM DRY DENSITY: 112.0 PCF

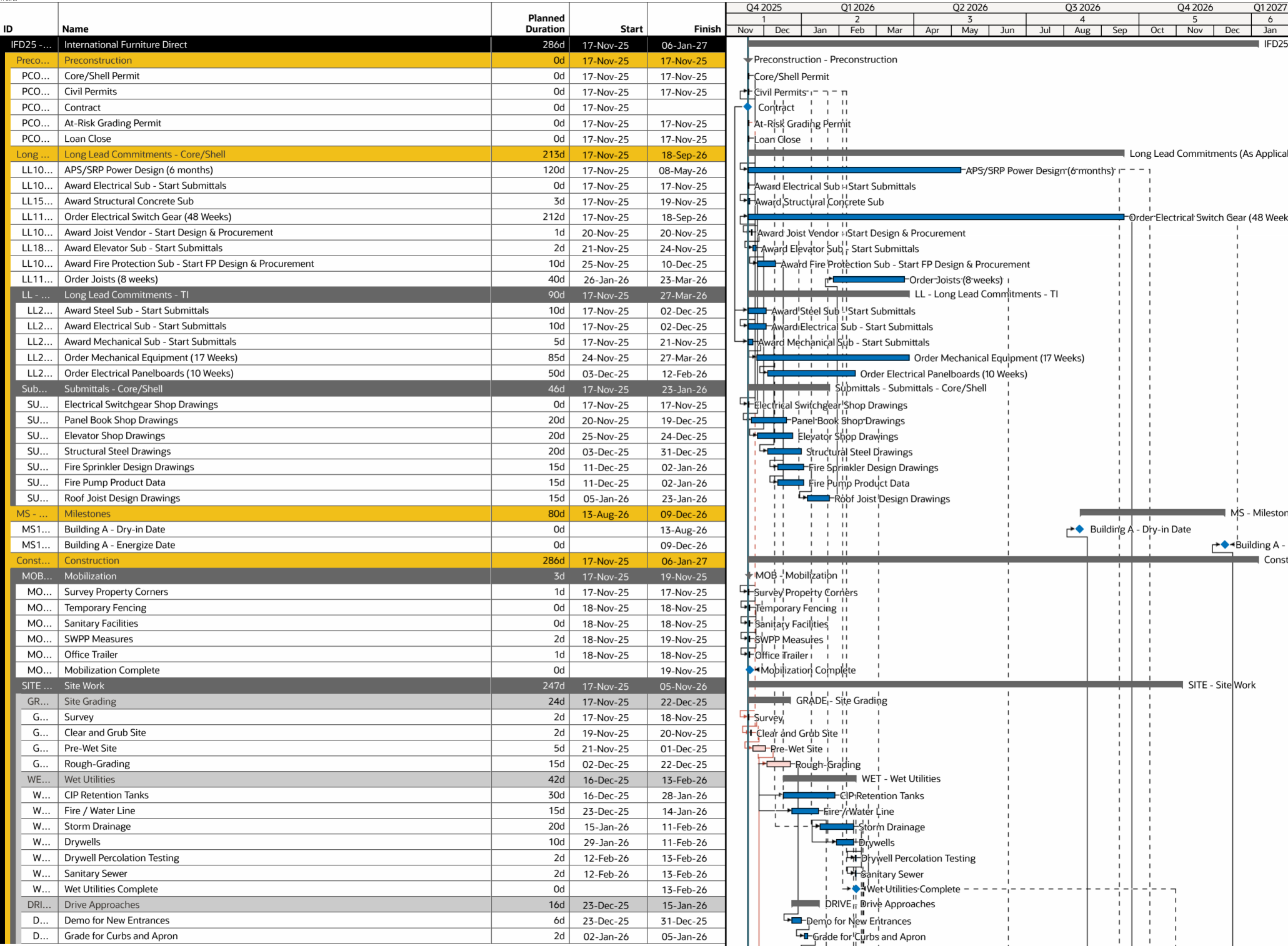
OPTIMUM MOISTURE CONTENT: 15.4%

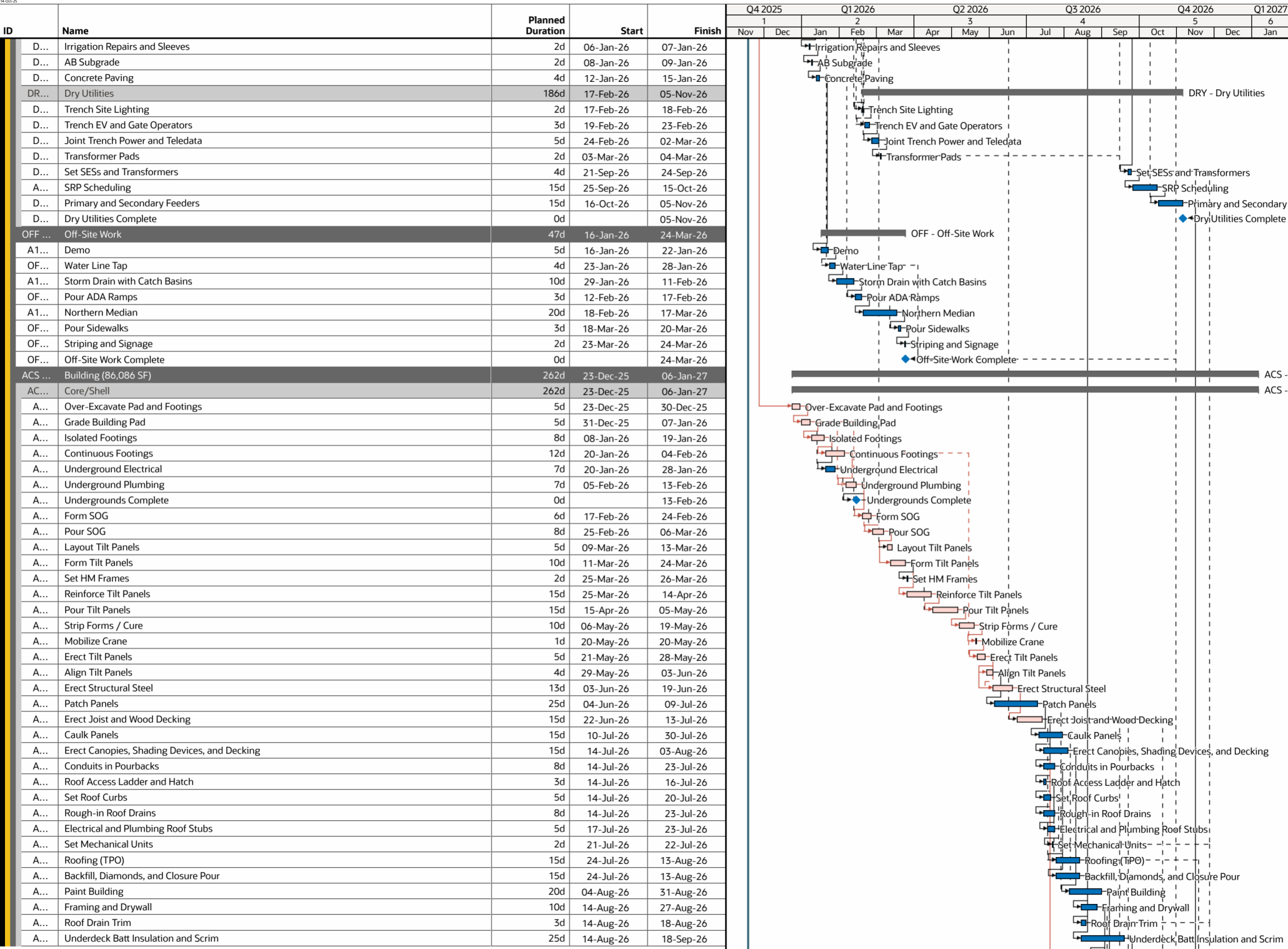


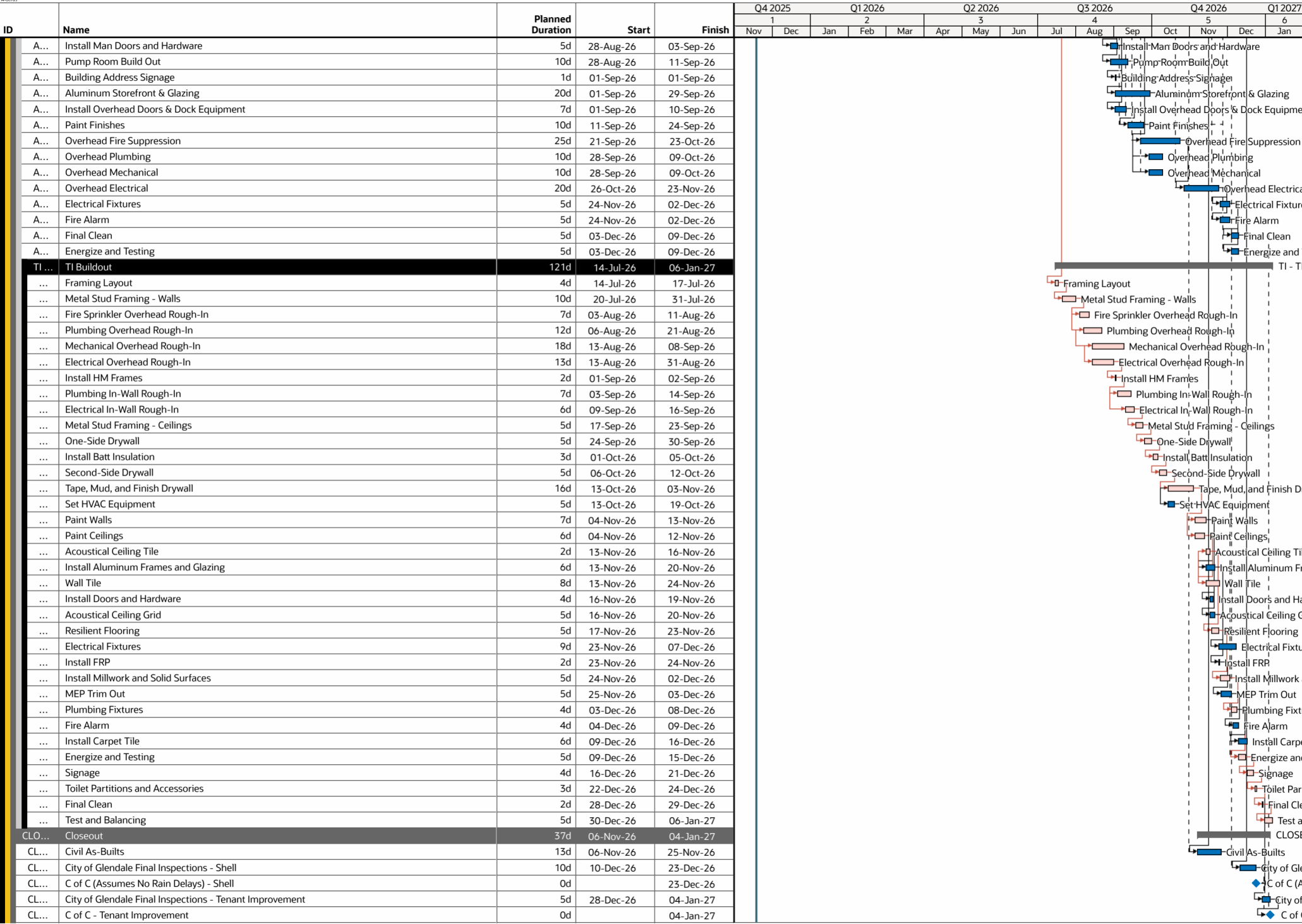
SWELL TEST DATA

BORING or TEST PIT No.	SAMPLE DEPTH, ft	MAXIMUM DRY DENSITY (pcf)	OPTIMUM MOISTURE CONTENT (%)	REMOLDED DRY DENSITY (pcf)	INITIAL MOISTURE CONTENT (%)	PERCENT COMPACTION	FINAL MOISTURE CONTENT (%)	CONFINING LOAD (psf)	TOTAL SWELL (%)
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B- 4, BS-2	5.0	112.0	15.4	106.8	13.2	95.3	21.4	100	2.8
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DESIGN BUILDER:

LGE Design Build

By:

Name:

Title:

Date:

SUBCONTRACTOR:

Desert Services

By:

Name: Kendra Ash

Title: Contract Manager

Date:

Subcontract has been reviewed by:

DS


Superintendent
Initial


Managing Director